

Annex IV — Industrial Field Operation and Governed Synchronization of the WR Graph™

BEAP™ / WR Desk™ / Optirando™ Canonical Specification — standalone Annex. Version 1.2 (see §Version History).

Document status: Normative unless explicitly marked as informative.

Applies to: WR Desk™ runtime, WR Graph™, Workflow Context Recorder (WCR), DeepFix sessions, BEAP™ transport (pBEAP™ and qBEAP™ classes), PoAC™-governed learning operations, and multi-device deployments comprising a central Host orchestrator and subordinate devices (Field Devices and network workstations).

Trademark notice: Optirando™, WR Desk™, WR Graph™, BEAP™, PoAE™, PoAC™, WRVault™, and WR Code® are used throughout as product and protocol designations of the Optirando™ system.

IV.0 Status, Scope, and Relationship to the Canon

This Annex specifies the industrial field operation of the WR Graph™: per-machine context regions, the taxonomy and governance of LoRA-class adapters in field deployments, the transport of LoRA-class adapters in BEAP™ packages, the synchronization of WR Graph™ regions between a central Host and authorized subordinate devices, WR Code® trigger classes and BEAP™ payload classes with their two distribution paths (public repository and direct message delivery), identity binding of internal handshakes, and multi-device session continuation.

This Annex does not redefine or extend wire formats, capsule serialization, cryptographic primitives, transport mechanisms, or PoAE™ execution semantics; these remain exclusively defined in the Canon and its Normative Annex. All enforcement mechanics of Annex I, the routing and gateway semantics of Annex II, the device-class and provisioning constraints of Annex II §§II.1.3 and II.6, and the acquisition governance of Annex III apply unchanged; nothing in this Annex weakens or contradicts them. Where this Annex refines the applicability of an existing rule to the intra-deployment case, the refinement is stated explicitly (see IV.5.7, IV.5.8, IV.5.10).

Relationship to the Canon (v202). The behavior specified here was introduced in the Canon in the sections "Optirando™ Station", "Dynamic WR Code® Visualization & DeepFix Sessions", and "From Fix to Knowledge: Building Verified Memories with WR Code®": per-machine WR Code® registration, DeepFix sessions over multimodal inputs, and the accumulation of a verified, semantically searchable per-machine knowledge memory. This Annex formalizes the mechanism behind that behavior. The terminology mapping in IV.1 is normative for interpretation: the concepts are continuous, and this Annex supersedes no conclusion of the Canon.

IV.1 Terminology Bridge to the Canon (Normative for Interpretation)

Canon (v202) term	This Annex / current Canon term
knowledge memory / knowledgebase (per WR Code®-registered asset)	WR Graph™ context region of that asset (IV.2)
allocated machine context, memory segments, historical logs	declarative and episodic dimensions of the asset's region
DeepFix session	DeepFix session (unchanged); its lifecycle is governed per IV.4
validated fix records with photos, logs, measurements	episodic evidence entities with provenance (Annex III §3)
Optrando™ Station (central deployment)	Host (IV.2.2; Annex II §II.1.3)
field tablet, smartphone at the machine (informal)	Field Device (Annex II §II.1.3; subordinate device, IV.2.2)
adapter (deterministic, versioned integration module of the Canon)	Integration Adapter (unchanged; executable, capability-declared)
— (not present in v202)	LoRA-class adapter (parametric artifact; IV.3)

IV.1.1 Adapter Disambiguation (Normative). The Canon reserves the unqualified term adapter for Integration Adapters: deterministic, versioned, executable integration modules with formal capability declarations, executed within an isolated adapter execution host. The artifacts specified in IV.3 are LoRA-class adapters: non-executable parametric artifacts (tensor weights). The two classes SHALL NOT be conflated. Conforming documents SHALL use the qualified terms wherever ambiguity is possible. Within Annex III, the unqualified term "adapter" denotes LoRA-class (parametric) adapters throughout; no Integration Adapter is governed there.

IV.2 Machine Sessions as WR Graph™ Context Regions

IV.2.1 One Graph, Region-Structured (Normative)

A deployment SHALL maintain one authoritative WR Graph™. Individual machines, assets, or processes SHALL NOT be represented as separate, isolated graphs.

Each WR Code®-registered machine or asset anchors a context region of the deployment graph. A region comprises the four knowledge dimensions for that asset:

- **Declarative** — manuals, configurations, SOPs, known failure modes, manufacturer documentation
- **Episodic** — this asset's execution and maintenance history, DeepFix sessions, solved-issue records, feedback events, outcomes
- **Procedural** — approved repair and maintenance workflows (WCR recordings), checklists, automation paths
- **Parametric** — LoRA-class adapter bindings applicable to this asset (IV.3), per Annex II §II.2.1

A WR Code® scan resolves the asset's region and, via cross-dimensional edges, simultaneously selects what is known, what has happened, which parametric competence applies, and which automations may be offered (Annex II §II.2.4).

Rationale (Informative). A single region-structured graph preserves fleet-level capabilities the Canon already claims: recurrence statistics across identical machine models, DeepResearch-style pattern analysis across machine fleets, and transfer of validated fixes between assets of the same class. Per-machine graph islands would forfeit these by construction.

IV.2.2 The Host and Subordinate Devices (Normative)

The Host is the deployment's primary orchestrator and trust root, in the sense of Annex II §II.1.3. The Host:

- SHALL hold the complete, authoritative WR Graph™ of the deployment, including all machine regions
- SHALL hold the deployment's internal session-template catalog (IV.6)
- SHALL be the exclusive point of graph mutation admission (IV.5.4); subordinate devices contribute evidence but never mutate the authoritative graph directly
- MAY serve multiple concurrent machine sessions and multiple subordinate devices

Subordinate devices are all devices paired under the Host's trust root. Two subclasses exist; both are subordinate in the identical governance sense and differ only in mobility and typical grant depth:

1. **Field Devices** (Annex II §II.1.3) — mobile or portable devices (smartphones, field tablets, smart-glasses units). They hold no independent trust root, no authoritative graph, no policy authority, and no consent authority; the exclusions of Annex II §II.6.4 apply in full.
2. **Network Workstations** — fixed devices within the deployment (e.g., a multi-screen analysis workstation). They are governed identically to Field Devices — paired, projection-only, subordinate — and typically hold deeper sync grants (IV.5.3). A Network Workstation gains no authority from being stationary.

Subordinate devices operate on projections (IV.5.3): grant-scoped subsets of the graph replicated for local, including offline, operation. No subordinate device SHALL establish handshakes, accept capsules, or conclude permission grants with external counterparties in its own right (Annex II §II.6.1); all external relationships belong to the Host.

Host and subordinate device are roles, not machines (Annex II §II.1.3). A single-node deployment holds both roles on one device, in which case the projection and grant machinery of IV.5 collapses trivially.

IV.2.3 Graceful Degradation (Normative)

Where a scan or query resolves no matching region — including assets not yet registered — the system SHALL degrade per Annex II §II.2.8: no adapter is loaded, no automation path is offered, no routing is fabricated, and the absence of domain knowledge for this asset is surfaced as such. A generic base-model answer MAY be given, clearly not attributed to asset-specific knowledge.

IV.3 LoRA-Class Adapters in Field Deployments

IV.3.1 Adapter Classes by Acquisition Path (Normative)

LoRA-class adapters in field deployments fall into exactly three classes, distinguished solely by acquisition path. All three are parametric artifacts in the identical technical sense; no class receives elevated authority.

1. **Field Adapters** — manufacturer- or OEM-provided domain adapters for specific device classes, machine families, or operational domains (e.g., a machine vision adapter for a specific PLC family, provided by its manufacturer).
2. **House-Internal Adapters** — adapters trained within the deployment on the organization's own material, including adapters trained on a specific machine's accumulated imagery, error states, and session evidence (IV.4).
3. **Optimization-Phase Adapters** — adapters produced by the Night Pipeline's data-set generation and training stage (Annex III §6.1, Stage 6) during the sorting and optimization phase of the WR Graph™.

House-Internal and Optimization-Phase Adapters differ only in initiating trigger (operator-initiated versus Night-Pipeline-proposed); their governance is identical.

Provenance-class mapping (Normative). The three classes map onto the provenance model of Annex III as follows:

- **Field Adapters are Acquired Capabilities** (Annex III §2). Each Field Adapter SHALL carry an acquisition evidence record per Annex III §3.8, with: publisher = the DNS-verified manufacturer identity (IV.3.4), signature and content hash per IV.3.2, base_model_binding per IV.3.2, declared_scope = the manufacturer's device classes, and channel = the manufacturer distribution path. Field Adapters are not regenerable by the deployment; their lifecycle follows Annex III §§3.8 and 6.3.
- **House-Internal and Optimization-Phase Adapters are Derived Capabilities** (Annex III §2). Each carries a full derivation chain; its data set is the persistent, provenance-bound asset and the adapter its regenerable compilation (Annex III §3.4).

The Source-entity acquisition enumeration of Annex III §3.3 is extended by the value manufacturer-provided. Accompanying manufacturer material (documentation, reference data sets, failure-mode catalogs) SHALL be recorded as Sources with this acquisition value; the adapter artifact itself is governed by its §3.8 acquisition evidence record.

IV.3.2 Serialization and Integrity (Normative)

- LoRA-class adapters SHALL be distributed and stored exclusively in a non-executable tensor serialization format (safetensors or an equivalent format that structurally excludes embedded executable code). Serialization formats capable of code execution on load (e.g., pickle-based checkpoints) are prohibited.
- Every adapter artifact SHALL be integrity-bound by content hash. Hash verification SHALL occur at admission and at every load. Verification failure fails closed.
- Every adapter SHALL be bound to an identified base model, a declared purpose or task class, and a defined policy scope (per Canon M.5.3). An adapter presented against a mismatched base model SHALL NOT load.

IV.3.3 Authority Constraints — Suggest Mode (Normative)

LoRA-class adapters:

- carry no operational authority. They operate strictly in Suggest Mode: their outputs are hypotheses, never decisions or actions.

- SHALL NOT expand capability scopes, influence policy evaluation, or substitute for any PoAE™ authorization requirement (Annex II §II.2.3).
- SHALL NOT be treated as authoritative content. Ground truth is resolved from the declarative and episodic dimensions of the graph; adapter outputs SHALL be reconciled against graph content and presented as interpretation, not fact (per Canon M.5.2: capability states, not authoritative content).

Authority remains exclusively with: orchestration logic, the WR Graph™'s deterministic routing (preparatory, never executive — Annex II §II.2.5), the policy layer, PoAE™ enforcement, and the authenticated operator.

IV.3.4 Context-Scoped Activation (Normative)

- Adapter activation SHALL be triggered exclusively by deterministic context resolution, which includes explicit operator selection — manual activation of an admitted adapter or selection of a mode carrying adapter bindings (Annex II §II.2.2). Operator selection does not bypass admission (Annex II §II.2.6, IV.3.7), does not relax Suggest Mode (IV.3.3), and is logged like any other lifecycle trigger.
- Field Adapters (manufacturer-provided) SHALL be bound at admission to the context regions of the providing manufacturer's device classes and SHALL NOT activate outside those regions — including under explicit operator selection. A manufacturer's parametric competence applies exactly where that manufacturer is already trusted through its equipment, and nowhere else. House-Internal and Optimization-Phase Adapters carry no such hard region restriction: an operator MAY manually activate an admitted House-Internal adapter in a compatible sibling context (e.g., another machine of the same model), consistent with the fleet-transfer rationale of IV.2.1; the activation is logged with the operator selection as the context reference.
- Manufacturer identity SHALL be verified through DNS-based publisher verification, reusing the verification procedure required of BEAP™ Gateway operators (Annex II §II.3.2). No separate manufacturer PKI is introduced.

Trust rationale (Informative). A trusted manufacturer already shapes diagnostics through firmware, PLC software, HMI software, and the hardware itself. A manufacturer-provided adapter therefore shares the manufacturer's trust anchor. It does not share the equipment's trust domain: firmware runs on the machine and its faults manifest in machine behavior, whereas an adapter runs inside the operator's reasoning system and its faults manifest as skewed suggestions, which are harder to observe. Context-scoped activation, Suggest Mode, provenance display (IV.3.5), and telemetry (IV.3.6) are the compensating controls that make manufacturer provisioning safe without pretending the two trust domains are identical.

IV.3.5 Provenance Display (Normative)

Whenever an adapter contributes to a displayed suggestion, diagnosis, or interpretation, the user interface SHALL make the contributing adapter's identity and class (Field / House-Internal / Optimization-Phase) inspectable at the point of suggestion. Adapter influence SHALL NOT be silent.

IV.3.6 Telemetry, A/B Baseline, and Retirement (Normative)

Per-adapter telemetry, idle-time A/B comparison against the bare base model, and retirement proposals apply unchanged per Annex III §6.3. Field Adapters are not exempt: a manufacturer adapter that no longer outperforms the base model — including after a base-model upgrade — SHALL be proposed for retirement as a Class-3 mutation.

Retirement semantics follow Annex III §6.3 by provenance class: House-Internal and Optimization-Phase Adapters remain regenerable from their retained data sets; Field Adapters are deactivated with the acquisition record retained as provenance, and the retirement proposal MAY include a manufacturer update check for a version compatible with the current base model — installing such a version is a new acquisition (Class 3, Annex III §3.8).

IV.3.7 Admission (Normative)

Admission of any LoRA-class adapter into the WR Graph™ — regardless of class — is a Class-3 mutation under PoAC™ (Annex III §4.4) and requires an explicit Consent Record, or coverage by an Organizational Authority Consent (IV.5.7) where deployment policy so provides. For Field Adapters, verification of the acquisition evidence record (Annex III §3.8) — publisher identity, signature, content hash, version, base-model binding — is a precondition of admission per Annex II §II.2.6, and re-verification SHALL be repeatable on demand; failed re-verification is an invalidated acquisition (Annex III §3.5). Until admitted, adapters remain in their libraries or in the sandbox trust domain and are unavailable to routing (Annex II §II.2.6).

IV.3.8 Adapter Transport in BEAP™ Packages (Normative)

LoRA-class adapters are transportable artifacts, not resident-only assets:

- A LoRA-class adapter MAY be carried as a payload artifact in a BEAP™ package of any class (pBEAP™ or qBEAP™), alone or alongside a session template — exclusively in the non-executable tensor serialization of IV.3.2 (safetensors-class), content-hash-bound and publisher-signed. An adapter travels over the network exactly like a template.
- **Class-dependent carriage rule.** Under pBEAP™ (public class), an adapter payload SHALL reference a repository-hosted artifact by content hash: the weights themselves reside in the public Optirando™ repository, publicly auditable per IV.6.3; a pBEAP™ package SHALL NOT carry unregistered adapter weights. Under qBEAP™ between parties holding an established handshake, adapter weights MAY be carried directly and confidentially, subject to the mandatory evidence annotation of IV.6.5. The asymmetry is deliberate: a recipient can inspect a text-pure template but cannot meaningfully audit a tensor blob, so binary competence is admissible only where it is either publicly auditable (repository) or bilaterally evidence-bound under an authenticated relationship (qBEAP™).
- In transit and at rest on receipt, adapter payloads SHALL be handled as encrypted, inert blob artifacts referenced by content hash (blob-inertness preserved); capsule text remains text-pure. An adapter payload is never rendered, never interpreted, and never loaded as part of package processing.
- **Transport confers no trust.** An adapter received in any package class is external-origin material regardless of transport encryption: it SHALL traverse the Ingestor–Validator pipeline (Annex I §I.3), SHALL NOT load before signature and hash verification against its acquisition evidence (Annex III §3.8) and its evidence annotation (IV.6.5) where delivered

over the direct message path, and SHALL NOT enter routing before Class-3 admission (IV.3.7). Encryption (qBEAP™) provides confidentiality and recipient binding — not admissibility.

- **Distinction from synchronization.** Carrying an adapter artifact in a BEAP™ package is artifact distribution under the Distribution Boundary — publication is Class 3 at the sender, acquisition is Class 3 at the receiver (Annex III §4.5). It is not WR Graph™ region synchronization and is not affected by the transport-class restriction of IV.5.2, which governs graph structure and region content, not individually published sealed artifacts. Intra-deployment projection of already-admitted adapter weights to subordinate devices remains governed by IV.5.6.
- **Gating summary.** In all cases: acquisition is PoAC™-gated, exercise is PoAE™-gated. A transported adapter, once admitted, operates in Suggest Mode like every other LoRA-class adapter (IV.3.3); nothing about its transport path expands capability scopes or pre-satisfies any authorization requirement.

Manipulation risk (Informative). The verification chain — publisher signature, content-hash binding at admission and at every load (IV.3.2), the mandatory evidence annotation for direct delivery (IV.6.5), and public auditability on the repository path (IV.6.3) — makes undetected alteration of adapter weights or their declared capability manifest infeasible in any deployment profile. Silicon Cargo™ (enterprise) and High-Assurance deployments additionally enforce mutual hardware attestation of endpoints for direct BEAP™ exchange (Canon, v202 ingress trust classes) and attestation-bound device identities (IV.7.5), closing endpoint substitution and tampered-environment risks around adapter exchange.

IV.4 The Field Loop as a PoAC™ Instance

IV.4.1 Principle (Normative)

The industrial field loop — capture, evidence, gap, remediation, adapter — is an instance of the PoAC™ acquisition discipline (Annex III), not a parallel mechanism. No element of the loop mutates the WR Graph™ without an authorized, auditable path.

IV.4.2 Session Capture (Normative)

During a DeepFix session at a machine, the operator MAY contribute, and the system MAY capture:

- photographs (components, damaged parts, HMI screens, error codes, log screens)
- audio (verbal problem description) and its transcription
- free text
- video and live vision input (including smart-glasses capture)
- direct machine logs and telemetry via declared API connectors
- structural signals where a software surface is involved

All captured material enters as external-origin input through the Ingestor–Validator pipeline (Annex I §I.3), is recorded as Source entities and evidence events with full provenance (Annex III §§3.3, 5.2), and is scope-bound to the asset's region.

Media originals (photos, audio, video) SHALL be handled as encrypted, inert blobs per the blob-inertness invariant and SHALL be referenced by content hash; graph entities themselves remain structured textual artifacts (text-purity preserved). Graph entities reference both the original (by its

content hash, custody: sandbox domain) and its reconstruction (by its own hash, with a provenance edge to the original).

Media Reconstruction and Analysis Placement (Normative). Media originals SHALL NOT be decrypted, decoded, or persisted within the Host trust root in any deployment tier. They SHALL be routed to the sandbox trust domain — on single-device deployments, the hypervisor-isolated sandbox VM co-resident on the host machine, using the platform's isolation primitive as defined by the Canon's deployment tiers (crosvm microVM on Linux, Hyper-V on Windows Pro, VirtualBox on Windows Home); on higher tiers, dedicated sandbox hardware — where the following pipeline applies:

1. **Reconstruction.** The sandbox decodes the original and re-encodes it with the deployment's own trusted encoder into a Reconstructed Media Artifact: perceptually identical to the original, structurally new — fresh container, fresh encoding, none of the original byte structure. Original metadata is not carried over binarily; declared fields (e.g. capture time) are extracted as separate text-pure artifacts.
2. **Delivery.** Only Reconstructed Media Artifacts and text-pure extracts leave the sandbox toward the Host. The Host AI — including its vision inference and any perception adapters (IV.3), whose verification and admission rules (IV.3.2, IV.3.7) apply unchanged — operates exclusively on reconstructions. The computationally light conversion runs on the sandbox hardware; the computationally heavy inference runs on the Host, over safe input.
3. **Custody.** The original remains in sandbox-domain custody as an encrypted, inert, hash-bound blob for evidence and audit purposes; it is never needed for analysis or display again.
4. **Fail-closed.** Media that cannot be reconstructed — unknown format, decode failure, structural anomaly — remains confined to the sandbox domain and SHALL NOT be forwarded in any form; the failure is logged as an evidence event.
5. In High-Assurance Mode, deployments SHOULD perform reconstruction in ephemeral converter instances that are reset after each batch, so that no parser compromise can persist. Isolation strength is platform-dependent; which platforms and tiers qualify for which assurance claims is governed by the Canon's tier model, not by this section.

Text visible within imagery survives reconstruction by design (the pixels are the content). It is evidence data: it SHALL NOT be interpreted as instruction, SHALL NOT influence routing state directly, and SHALL NOT satisfy any authorization requirement.

Rationale (Informative). Media decoders are historically the most exploit-prone parser class in computing. Reconstruction removes the byte structure that format-level exploits, polyglot files, and malicious metadata depend on: only the pixels cross the trust boundary, never the bytes.

Reconstruction does not, and cannot, sanitize pixel-carried semantics — hence the instruction prohibition above. This mechanism generalizes the Canon's Reconstructed Artefact model from documents (deterministic text representations) to media.

IV.4.3 Session Feedback (Normative)

At session end, the operator's confirmation or rejection of the outcome SHALL be persisted as a Feedback Event per the schema of Annex III §5.2, with target_ref = the DeepFix session, project_ref = the asset's region, verdict, and optional commentary. This event is the primary evidence signal driving the asset's graph evolution.

IV.4.4 Evolution of the Region (Normative)

Accumulated evidence drives improvement through the Night Pipeline (Annex III §6) with the escalation ladder unchanged: workflow compilation → prompt profile → RAG enrichment → validator → tool integration → external acquisition (where a channel is configured) → adapter training (Annex III §6.2). In particular:

- Recurring fixes SHALL surface as process gaps, proposing WCR recordings of the validated repair procedure (Annex III §5.4, §5.7).
- Before adapter training is proposed for a perception or competence gap on an asset class, the Pipeline SHALL, where a manufacturer channel is configured, check for and prefer a suitable, verifiable Field Adapter as the external-acquisition rung (Annex III §5.4): acquiring the manufacturer's validated competence for its own device class is cheaper and better-evidenced than training a local equivalent. Acquisition remains Class-3, consent-gated, with full §3.8 acquisition evidence.
- Where evidence demonstrates that cheaper rungs — including external acquisition where available — cannot close a perception gap for an asset (e.g., visual identification of failure states specific to this machine), the Pipeline MAY propose data-set generation from the asset's accumulated session imagery and subsequent training of a House-Internal or Optimization-Phase adapter. Both steps are Class-3, consent-gated, with the data set as the persistent, provenance-bound asset and the adapter as its regenerable compilation (Annex III §3.4).
- Baseline checks apply: commodity knowledge already mastered by the base model SHALL NOT generate proposals (Annex III §5.3).

Operational effect (Informative). Over time, the asset's region converges on precise, efficient handling of recurring machinery issues. Because the region — not any individual — holds the validated fixes, the deployment retains diagnostic capability across shift changes and staff turnover: a technician on the next shift, or a newly onboarded one, scans the same WR Code® and inherits the full validated history, procedures, and trained perception for that machine.

IV.5 Graph Region Synchronization (Host ↔ Subordinate Devices)

IV.5.1 Context-Delivery and Computation-Placement Modes (Normative)

Three modes exist; their selection is deployment- and use-case-dependent, per the placement principle that knowledge and computation are placed where they create the best trade-off between performance, trust, latency, privacy, and offline capability.

1. **Host-served operation (enterprise default).** The authoritative WR Graph™ and the computation remain entirely on the Host. The subordinate device acts as capture and display surface: it transmits the query and captured session inputs up over qBEAP™; the Host performs context resolution against the full graph, adapter selection and loading, and inference; the results — suggestions, resolved procedures, reconciled history — are looped back to the device. No graph content persists on the device beyond transient session display state. Host-served operation requires connectivity and yields the full depth of the

deployment graph at every interaction; it is the normative default for enterprise deployments.

2. **Persistent region projection (offline-capable, internal-only).** Within a single deployment's trust domain, an authorized subordinate device MAY hold a persistent projection of designated graph regions, maintained through synchronization roundtrips (IV.5.4). Projections are provisioned content in the sense of Annex II §II.6.2 and SHALL satisfy its sealing, version-stamping, and lease requirements (IV.5.9, IV.5.10).
3. **Capsule-carried context (counterparty default, unchanged).** BEAP™ capsules carry the memory and context required for their task — global memory, agent memory, session context. This remains the general mechanism for exchange with counterparties, for both pBEAP™ and qBEAP™. No counterparty requires, and none SHALL receive, the WR Graph™ itself.

A device MAY combine modes 1 and 2: host-served while connected, falling back to its granted projection when disconnected. Mode selection per session SHALL be deterministic from the device's grant and its connectivity state, and SHALL be logged. Nothing in mode selection alters authorization: consequential steps are gated per IV.8.3 in every mode.

IV.5.2 Transport Class Restriction (Normative)

- Graph region synchronization SHALL be carried exclusively over qBEAP™ under an established internal handshake (IV.5.3, IV.7).
- A capsule carrying WR Graph™ structure or content over pBEAP™, or under a public or asymmetric handshake, is a protocol violation and SHALL fail closed at both endpoints.
- This rule is the internal-boundary counterpart of the cloud abstraction boundary (Annex III §7): WR Graph™ content crosses neither the learning-assist cloud boundary (Annex III §7.3) nor the public-handshake boundary — two edges of the same invariant. Publication of an individual artifact to an external party is neither of these: it occurs, if at all, only through the Distribution Boundary as a separate, consent-gated act (Annex III §§4.5, 7.1), never as region synchronization.

IV.5.3 Sync Grants and Projections (Normative)

The internal handshake between Host and a subordinate device SHALL carry a sync grant (the internal application of the permission-grant model, Annex II §II.3.1), specifying:

- the granted regions (machines, assets, sites, or region groups)
- optionally, the granted dimension depth per region (e.g., full depth for Network Workstations; declarative + procedural + parametric with condensed episodic history for a field tablet)
- validity period and revocation conditions
- a lease: the maximum offline validity period of the grant and its projection, per Annex II §II.3.1 (see IV.5.10)

Projection SHALL be derived exclusively from the sync grant. Content outside the grant SHALL NOT be transmitted; a request outside the grant SHALL fail closed. Default is deny.

IV.5.4 Roundtrip Model (Normative)

A synchronization roundtrip comprises three stages, reusing the versioned, hash-addressable context-block model of the Relational Context Graph (Annex I, Local Persistence and Incremental Enrichment):

1. **Inventory.** The device transmits its per-region version cursor or block-hash inventory for granted regions.
2. **Delta (down-sync — replication).** The Host responds with graph-sync capsules containing only the missing or updated blocks for granted regions, at granted depth. Down-sync is replication of already-admitted graph content; it is not a new admission.
3. **Contribution (up-sync — evidence).** The device transmits session output as ordinary evidence capsules: episodic events, captured media references, logs, and Feedback Events from DeepFix sessions.

Deltas follow the append-and-enrich model: blocks are created once, referenced by (block_id, hash), and enriched incrementally; full-payload retransmission is avoided by design. The per-region version cursor constitutes the version-stamping of the projection required by Annex II §II.6.2.

Every roundtrip SHALL include a lease renewal check (IV.5.10): renewal verifies grant validity and revocation state before any delta is exchanged.

IV.5.5 Host Authority and the Absence of Merge (Normative)

- The Host's graph is authoritative. Subordinate devices SHALL NOT mutate the authoritative graph directly.
- Up-synced contributions SHALL traverse the Host's Ingestor–Validator pipeline (Annex I §I.3) and are admitted under PoAC™ classification like any other material.
- There is no bidirectional graph merge. Conflicting field observations are not a merge conflict: they are distinct episodic events, both retained with provenance and temporal ordering. Contradiction handling follows bi-temporal invalidation (Annex III §3.7), performed on the Host.

Security effect (Informative). A compromised or lost field device cannot poison the authoritative graph directly; at worst it submits malformed or malicious evidence, which is precisely the input class the ingestion pipeline exists to contain.

IV.5.6 Payload Classes (Normative)

- Graph deltas are structured textual artifacts (text-purity preserved).
- Media referenced by synced entities travels as Reconstructed Media Artifacts (IV.4.2), hash-bound and encrypted in transit; originals remain in sandbox-domain custody at the Host site and are not distributed. Devices MAY receive reconstructions lazily (on demand) within their grant; blob-inertness is preserved on every device.
- Adapter weights (parametric dimension) MAY be projected to devices whose grant covers the corresponding regions, so that offline field diagnosis with the asset's trained perception is possible. Weights sync under IV.3.2 integrity rules, with hash verification at every load on the receiving device. Intra-deployment replication under an internal handshake occurs within a trust domain of the deploying identity in the sense of Annex II §II.2.3; the device inherits the Host's admission consent (IV.5.7) and SHALL NOT require re-consent per device.

IV.5.7 PoAC™ Classification of Synchronization (Normative)

Synchronization operations map onto the assignment table of Annex III §4.5 as follows:

- **Down-sync of already-admitted content is provisioning** under Annex III §4.5, not a new acquisition. Provisioning of non-confidential admitted content to a granted subordinate device is Class 2 (autonomous, reported, reversible by recall or expiry). Provisioning of content classified as confidential — which, in industrial deployments, region projections ordinarily are — is Class 3 and SHALL be covered by an **Organizational Authority Consent**: a persisted, standing Consent Record, issued under the deployment's policy authority (the policy-gate mechanism of Annex III §3.6), that covers a declared class of synchronized content for declared device classes and grant depths. Issuance, modification, and revocation of an Organizational Authority Consent are themselves Class-3 policy mutations and are logged accordingly.
- **Up-sync contributions classify normally** under Annex III §4: new Sources and evidence are Class-3 or covered by a standing Organizational Authority Consent for the declared evidence class; Class-1 linking and condensation over admitted events proceeds per the assignment table.
- **Lease renewals** performed during roundtrips are Class 1 (Annex III §4.5).

Nothing in this section weakens Annex III §4.4: no content enters the authoritative graph without a Consent Record, and no confidential content leaves the Host without one. This section defines which Consent Record covers replication and routine field evidence, so that per-delta operator consent across a device fleet is not required.

IV.5.8 Intra-Deployment Evidence and Routing State (Normative Refinement of Annex II §II.2.5)

Annex II §II.2.5 derives routing state exclusively from locally recorded, policy-authorized signals and excludes external signals — including incoming capsules — from routing state. For the purposes of that rule, devices operating under the same organizational authority, over internal (qBEAP™) handshakes established per this Annex, are not external parties: their up-synced evidence constitutes local, policy-authorized signal once admitted through the Host's ingestion pipeline. Evidence arriving over public or asymmetric handshakes, from publishers, or from any party outside the deployment's organizational authority remains external and SHALL NOT influence routing state, unchanged.

IV.5.9 Sealing, Revocation, and Projections at Rest (Normative)

- Projections at rest SHALL be sealed in the sense of Annex II §II.6.2 by two independent, cumulative bindings: **device-identity binding** — projection material is cryptographically bound to the enrolled device identity (IV.7.3) and unusable on any other device — and the **custody invariant** — projection storage keys derive from the authenticated operator session (IV.7.4); without a valid, second-factor-authenticated session, the projection is cryptographically inert.
- Revocation of a device's internal handshake SHALL immediately prevent all future synchronization roundtrips under it (consistent with Annex I §I.4.2), and SHALL take effect against the projection no later than lease expiry (IV.5.10).

- In High-Assurance Mode (Annex I §I.8), deployments SHALL additionally support a **wipe-on-next-contact** policy: a revoked device that reconnects receives a destruction directive for its projection before any other exchange, and the wipe result is logged.

IV.5.10 Lease Semantics for Offline Projections (Normative)

Sync grants are permission grants; the lease requirement of Annex II §II.3.1 applies to them in full:

- Every sync grant SHALL carry a lease — a maximum offline validity period appropriate to the device class and content confidentiality (deployment policy; e.g., shift-length leases for field tablets, longer leases for Network Workstations on the internal network).
- A projection whose lease expires without renewal SHALL become unusable on the device: expiry is fail-closed, rendering the projection cryptographically inert pending renewal. Expiry does not delete evidence captured on the device; captured contributions remain queued for up-sync at next contact.
- Renewal occurs only through contact with the Host and SHALL include a revocation check and a version check (Annex II §II.6.3): a projection prepared against a superseded graph state is refreshed by the ordinary delta mechanism (IV.5.4) as part of renewal.
- The lease and the custody invariant (IV.5.9) are independent, cumulative protections: the custody invariant defeats device theft and loss of logged-out devices at any time; the lease bounds the revocation latency for a device that remains authenticated but disconnected. Revocation therefore takes effect immediately where connectivity exists and no later than lease expiry where it does not.

IV.6 WR Code® Classes, BEAP™ Payloads, and Distribution Paths

IV.6.1 WR Code® Classes (Normative)

A WR Code® is a trigger: its scan initiates a BEAP™ exchange. It is not itself a template, an artifact, or a hosting class. WR Code®s exist in exactly two classes, unchanged from the Canon:

1. **Public WR Code®s.** Resolvable by anyone. Resolution occurs against the public Optirando™ repository (optirando.com) over pBEAP™.
2. **Internal WR Code®s.** Resolvable only by enrolled devices of the deployment's organization. Resolution occurs against the Host over qBEAP™ under an internal handshake. An internal WR Code® scanned by a non-enrolled device SHALL resolve to nothing — silent non-acceptance, never an error distinguishable from non-existence, and never a fallback to public resolution.

Resolution class is determined deterministically from the code's registration; each code belongs to exactly one class and SHALL NOT be resolvable through both paths.

Where a public WR Code® carries a handshake invitation, its scan additionally participates in the fast public initiation class governed by Annex IX (public_bearer invitation, claim, consent, and first-interaction semantics). This participation does not alter the resolution rules of this section: resolution class remains deterministic from registration, single-path, and fail-closed; internal WR Code®s never participate in public initiation.

IV.6.2 BEAP™ Payload Classes (Normative)

A BEAP™ message initiated by a scan — or exchanged directly under a handshake without any scan — MAY carry, alone or in combination:

- a **session template** (automation) — orchestration structure, agent roles, WCR-recorded workflows
- **context blocks** — the knowledge the session needs
- **capability declarations and capability requests**
- **LoRA-class adapters** — parametric competence, exclusively per IV.3.8 (safetensors-class, hash-bound, publisher-signed)

Together these deliver the specialized AI to the recipient in a secure format: the automation, its context, its declared capabilities, and its trained competence, in one governed exchange. The artifact taxonomy (structure, capability declarations, stamping) is identical regardless of payload combination or distribution path; paths differ exclusively in visibility and handshake class (IV.6.3). No payload class is executable in transit; all payloads traverse the Ingestor–Validator pipeline on receipt, and adapters additionally follow IV.3.8.

IV.6.3 Two Distribution Paths (Normative)

1. **Public repository path (pBEAP™).** Artifacts published for public WR Code® resolution are hosted in the public Optirando™ repository. Everything on this path — templates, identifiers, capability declarations, and adapter artifacts (safetensors) — SHALL be publicly inspectable and auditable. Sealed or encrypted content SHALL NOT be hosted in the public repository.

Rationale (Informative). Public auditability is the structural security control of the public path, not a convenience: a public WR Code® is scannable by any device, so everything a scan can deliver must be inspectable by anyone before and after delivery. Combined with non-executable serialization (IV.3.2), the ingestion pipeline, and consent-gated admission, this is what prevents the public scan path from becoming an infection vector — which would otherwise make the system a massively attractive target.

2. **Direct message path (pBEAP™ / qBEAP™).** Artifacts MAY equally be delivered directly between parties as BEAP™ messages under an established handshake — including confidentially over qBEAP™ between distinct organizations (enterprise-designated handshakes where applicable, Canon A.3.054.4). Templates, adapters, capability declarations, and context can thus be exchanged confidentially: content end-to-end encrypted, recipient-bound, invisible to third parties, never touching the public repository.

On this path, public inspectability is absent by design. The compensating controls are therefore mandatory: every directly delivered artifact SHALL carry the evidence annotation of IV.6.5, SHALL traverse the receiving side's ingestion and verification pipeline, and SHALL be admitted only as a Class-3 acquisition under PoAC™ (Annex III §§3.8, 4.4). In enterprise (Silicon Cargo™) and High-Assurance deployments, PoAC™ discipline on this confidential delivery path is essential: it is the sole auditability substitute for the public inspection the path deliberately forgoes, complemented by mutual hardware attestation of endpoints for direct exchange.

IV.6.4 Internal Template Governance (Normative)

Internal templates — hosted on and loaded from the Host for the deployment's own enrolled devices — SHALL carry governance metadata binding at minimum: version, author identity, approval state, capability references, integrity verification (WRStamp), and organizational scope. Version, author, capabilities, integrity, and lineage are already bound by stamped code capsules (Annex I §I.7.2); approval state and organizational scope are the additive fields.

- Organizational scope SHALL restrict resolution and execution to devices enrolled in the deployment (holding a valid internal handshake and, where required by policy, attestation).
- Admission of an internal template into the Host catalog, and every revision, is a Class-3 mutation under PoAC™ (Annex III §4.4), subject to the WCR authoring and validation lifecycle where the template embeds recordings (Annex II §II.5). Background-proposed revisions follow Annex II §II.4: proposed, never applied silently.
- Organizational scope is recipient-binding at group granularity. It composes with, and never weakens, per-capsule recipient binding.

Artifacts received over the direct message path from another organization are external-origin material regardless of transport encryption: admission into the receiving deployment's catalog is a Class-3 acquisition carrying acquisition evidence per Annex III §3.8, with publisher = the originating organization's DNS-verified identity, verified against the artifact's evidence annotation (IV.6.5). Embedded recordings remain subject to the ingestion, validation, and sandbox-confinement rules of Annex I and Annex II §II.5.4; embedded adapter payloads follow IV.3.8.

IV.6.5 Mandatory Evidence Annotation for Directly Delivered Artifacts (Normative)

Every artifact delivered over the direct message path — templates, LoRA-class adapters, capability declarations — SHALL be hashed and annotated at dispatch with, at minimum:

- the publisher identity (DNS-verified, per Annex II §II.3.2 verification)
- the content hash of the artifact
- the version
- the declared capability manifest (for templates) or the declared scope and base-model binding (for adapters), hash-bound
- a dispatch timestamp

Both parties SHALL retain the evidence annotation, bound to the handshake under which delivery occurred. The purpose is bilateral provability: in case of a dispute, malfunction, or security incident, either party can independently prove that a given template, adapter, capability declaration, or context artifact existed, in which exact version and content, from which publisher, at what time — without either party being able to repudiate or silently substitute it after the fact.

The annotation MAY additionally be registered as a public hash-commitment record with an anchored timestamp (e.g., blockchain-anchored per the Canon's provenance mechanisms); a hash commitment discloses no content, so confidentiality of substance is preserved while existence and integrity become provable to any third party. Public anchoring SHOULD be used in enterprise (Silicon Cargo™) and High-Assurance deployments.

Recipient verification SHALL compare the received artifact against its evidence annotation before admission; a mismatch is a failed acquisition verification and fails closed (Annex III §§3.5, 3.8).

IV.7 Identity Binding and Cryptographic Hardening of Internal Handshakes

IV.7.1 Layered Identity Model (Normative)

Internal deployments operate three identity layers. The layers are complementary; none substitutes for another.

1. **Endpoint identity** (inter-party, unchanged): anchored to a confirmed email address; enterprise-anchored endpoint identities additionally require DNS-verified domain control and confirmed organizational address ownership (Canon A.3.054.3). This layer provides identity proofing at establishment — it determines with whom a handshake is concluded. It provides no runtime protection of handshake material.
2. **Operator identity** (intra-deployment): the operator's SSO identity at the deployment's identity provider — the stable subject identifier and realm — with mandatory second-factor enforcement at session login to the orchestrator.
3. **Device identity**: the device's registered asymmetric key material (orchestrator device keys), bound at enrollment. In High-Assurance Mode, device identity keys are attestation-bound (IV.7.5).

IV.7.2 Threat Statement (Informative)

Without the mechanisms of IV.7.3–IV.7.5, an established handshake degenerates into a pure possession credential: whoever obtains the handshake material can exercise the relationship, and the establishment-time email anchoring contributes nothing at the moment of use. This Annex therefore treats endpoint anchoring as necessary but insufficient, and adds custody protection (at rest), derivation binding (in use), and attestation binding (environment integrity) as independent, cumulative hardening layers.

IV.7.3 Cryptographic Derivation Binding (Normative)

- The handshake identifier and all handshake-derived eligibility material for internal handshakes SHALL be derived over, at minimum: the sender and receiver endpoint identities (per Canon A.3.054.3), the operator identity (stable subject identifier and realm), the device identity (device public key), and the handshake context.
- Verification at either endpoint SHALL fail closed where the presenting party cannot prove the bound operator identity and device identity. Consequently, handshake material exercised from a different operator identity, or from a different device, fails cryptographic verification at the counterparty — not merely a local policy check.
- Binding SHALL be to stable identifiers only (subject identifier, realm, device public key) — never to session tokens, access tokens, or other rotating session artifacts. Rotating artifacts gate use (IV.7.4); stable identifiers define whose handshake it is.
- Sync grants (IV.5.3) and Organizational Authority Consents (IV.5.7) SHALL be bound to the same identifiers.

IV.7.4 Session Gating and Custody Invariant (Normative)

- Use of a bound handshake or grant SHALL require a live, second-factor-authenticated operator session at the deployment's identity provider. Second-factor enforcement SHALL be mandatory at the identity provider (a required authentication step, not a user-optional setting).
- Handshake material, sync-grant records, and projection storage keys on any device SHALL reside in vault storage whose keys derive from the authenticated operator session and are bound to the enrolled device identity (IV.5.9). Consequently: possession of a device, or exfiltration of its storage (including the handshake ledger), does not yield usable handshake material absent a valid second-factor-authenticated session; and logout renders handshake material and projections cryptographically inert on that device until the next authenticated login.

IV.7.5 Attestation Binding (High-Assurance, Normative)

In High-Assurance Mode (Annex I §I.8):

- Device identity keys SHALL be generated within, and certified by, a hardware-attested execution environment (e.g., TPM- or TEE-resident keys with attestation evidence), such that the device identity itself carries proof of environment integrity.
- Handshake establishment SHALL verify the attestation evidence of the device identity key; the attested key enters the handshake derivation per IV.7.3, transitively binding the handshake to environment integrity. Material presented from a non-attested or tampered environment fails cryptographically.
- Attestation SHALL be re-validated per deployment policy (on interval or on demand, consistent with Canon A.3.054.4's dynamic verifiability). Re-validation failure SHALL suspend use of all handshakes bound to the affected device identity until attestation is restored.

IV.7.6 Coverage and Residual Risk (Informative)

Each layer defeats a distinct attack class: endpoint anchoring defeats establishment-time impersonation; the custody invariant defeats offline theft and loss of logged-out devices; derivation binding defeats replay of exfiltrated material under a different identity or from a different device, including insider misuse of another operator's material; attestation binding defeats use from integrity-compromised environments; the lease (IV.5.10) bounds the exposure window of a revoked but disconnected device. Not covered by identity binding: an adversary executing on a device during a live authenticated session acts with the operator's identity. This residual class is addressed by the enforcement architecture, not by handshake identity: per-step PoAE™ gating of consequential actions, the mandatory sandbox model (Annex I §I.2), and attestation re-validation.

IV.7.7 Enterprise Designation Compatibility (Normative)

Where a deployment operates Enterprise Handshakes (Canon A.3.054.4: hardware-attested orchestrators, enterprise-anchored endpoint identities), the requirements of this section are additive. Nothing in this Annex relaxes attestation or endpoint-anchoring requirements.

IV.7.8 Revocation Path (Normative)

Handshakes established under this Annex MAY be long-lived, including without scheduled expiry. Absence of expiry SHALL NOT mean absence of revocation; and absence of expiry SHALL NOT exempt a grant from the lease requirement of IV.5.10, which bounds offline usability independently of handshake lifetime. The following SHALL each independently terminate usability of a handshake and all grants under it:

1. explicit handshake revocation at the Host (per Annex I §I.4.2 semantics: future activation denied, historical integrity preserved) — effective immediately where connected, and no later than lease expiry on a disconnected device (IV.5.10);
2. disablement or deletion of the operator identity at the identity provider — no authenticated session can be established, vault keys become underivable, and handshake material is rendered inert (IV.7.4); additionally, derivation binding (IV.7.3) invalidates any attempt to re-bind the material to a substitute identity;
3. revocation of the device's enrollment (device identity), with wipe-on-next-contact where High-Assurance policy applies (IV.5.9).

IV.8 Session Continuation and Multi-Device Operation

IV.8.1 Session State as Synced Content (Normative)

DeepFix session state — captured inputs, intermediate findings, adapter suggestions with provenance, pending confirmations — SHALL be representable as region-scoped content subject to the synchronization model of IV.5. A session opened on one device MAY, within the operator's grant, be continued on another enrolled device after a sync roundtrip.

IV.8.2 Quick and Deep Sessions (Informative)

A smartphone or field tablet suffices for quick sessions at the machine: scan, capture, receive suggestions, confirm, rate. For complex machinery, the operator continues the same session at a multi-screen Network Workstation — full graph depth, side-by-side comparison of history and hypotheses, full documentation review. The continuation is the same session against the same region; no re-capture and no re-entry of context occurs. This mirrors, and formalizes, the Canon's capture-in-the-field / process-after-sync pattern.

IV.8.3 Consequential Steps (Normative)

Session continuation never carries authorization across devices implicitly. Every consequential step remains gated at its applicable PoAE™ tier on the device where it is confirmed; a confirmation pending on device A is not satisfiable by mere session arrival on device B.

PoAE™ gating authority resides with the Host (Annex II §II.3.5). A consequential step confirmed on a subordinate device SHALL be gated by either (a) live Host gating at confirmation time, or (b) a delegated enforcement component explicitly provisioned by the Host to that device and bounded to a declared tier ceiling; delegated enforcement components fall within the provisional Field Device scope (Annex II §§II.1.3, II.3.5). Absent connectivity and absent an applicable delegation, a consequential step SHALL halt — retained as a pending confirmation in session state — until Host contact is available. Suggestion, capture, and non-consequential session activity remain fully available offline within the projection.

IV.9 Worked Example — Field Technician at a Registered Machine (Informative)

A technician on the late shift finds packaging machine M-747 halting intermittently. She scans its WR Code® with a field tablet. The tablet holds a projection of her site's regions under a sync grant with a shift-length lease, renewed automatically at her last dock contact.

Resolution. The scan resolves M-747's context region. The graph selects across dimensions simultaneously: declarative (the machine's manual, its configuration, known failure modes for this model), episodic (M-747's own history — the same symptom occurred twice, four months apart), procedural (the approved inspection workflow recorded after the last occurrence), parametric (the House-Internal adapter trained, with consent, on M-747's accumulated session imagery).

Capture. She photographs the HMI error screen, the drive assembly, and a worn coupling; dictates a description; pulls the last 200 controller log lines over the declared API connector. Her photos are converted in the sandbox domain into reconstructed images (IV.4.2); the Host — and every colleague who later opens this session — only ever sees the reconstructions, never the original files. Everything enters through ingestion as provenance-bound evidence, scoped to M-747's region.

Interpretation. The adapter — identified as such on screen, with its class and lineage inspectable — flags the coupling wear pattern as matching the failure state from the prior episodes. The graph reconciles: episodic records show the previous validated fix, including the photos the day-shift technician attached back then. The system proposes the recorded repair workflow.

Governed execution. The workflow runs to the authorization boundary. Ordering the replacement part is a consequential step; it halts at its PoAE™ tier until she confirms — gated live by the Host over the plant network. Nothing the adapter suggested, and nothing the routing surfaced, pre-satisfied that confirmation. Had she been out of coverage, the confirmation would have remained pending in the session until contact, while capture and diagnosis continued offline within her lease.

Feedback. The fix works. She confirms; a Feedback Event with positive verdict and a short comment is persisted against the session and M-747's region.

Evolution. Overnight, the Night Pipeline links the event, updates confidence on the validated fix, and notes the recurrence. Because this failure has now occurred three times, it proposes (Class 3, in the Morning Digest, with the recurrence evidence attached) a preventive-maintenance workflow recording. The maintenance lead accepts; the procedure enters the region with full provenance.

Continuity. A week later, a newly hired technician on a different shift scans the same code. He has never seen this machine. He inherits everything: history, validated fixes, the trained perception, the new preventive procedure. If he scans an unregistered machine instead, the system says so — it offers no fabricated confidence, loads no adapter, and answers only from the base model, clearly marked as non-asset-specific.

IV.10 Relation to Prior Work and Claimed Contribution (Informative)

The constituent mechanisms — per-asset knowledge bases, maintenance-record retrieval, edge anomaly detection, LoRA adaptation, dataset-driven fine-tuning, device synchronization — exist

independently in prior work and in the Canon itself. The contribution claimed by this Annex is the normative composition: per-asset context regions of a four-dimensional graph, whose evidence-driven evolution (including the training of asset-specific parametric competence from field-captured material, and the consent-gated acquisition of manufacturer competence for the manufacturer's own device classes) is governed end-to-end by PoAC™ acquisition discipline, whose replication to subordinate devices rides exclusively on the deployment's existing qBEAP™ enforcement stack under grant-scoped, lease-bound, device-sealed projection with host-exclusive admission authority, and whose every suggestion remains subordinate to PoAE™ execution gating. Capability acquisition, capability distribution, and capability exercise are placed under one continuous proof discipline.

§Version History

Version	Date	Change
1.0	2026-07-08	Initial normative specification: terminology bridge to Canon v202 (knowledge memory → WR Graph™ context regions), region model and Host definition, LoRA-class adapter taxonomy (Field / House-Internal / Optimization-Phase) with serialization, authority, activation-scoping, provenance-display and telemetry rules, field loop as PoAC™ instance, qBEAP™-exclusive graph region synchronization with sync grants, host-authoritative contribution model and Organizational Authority Consent, internal (qBEAP™) template hosting with organizational scope, layered identity binding with custody invariant and revocation paths, session continuation, worked example.
1.1	2026-07-09	Alignment with Annex II v1.1 and Annex III v1.1. Device classes harmonized with Annex II §II.1.3: subordinate-device umbrella (Field Devices and Network Workstations), roles-not-machines statement, external-relationship prohibition, Annex II §II.6.4 exclusions applied (IV.2.2). Terminology bridge extended (Field Device row; Annex III adapter-usage note in IV.1.1). Provenance-class mapping added: Field Adapters as Acquired Capabilities with Annex III §3.8 acquisition evidence; House-Internal and Optimization-Phase as Derived Capabilities (IV.3.1); admission preconditioned on acquisition-evidence verification with fail-closed re-verification (IV.3.7); retirement semantics split by provenance class with manufacturer update check (IV.3.6). IV.3.4 activation triggers extended to explicit operator selection (manual adapter activation, mode selection) with admission, Suggest Mode, and logging preserved; Field Adapter region binding made explicitly selection-proof; manual sibling-context activation of House-Internal adapters permitted per the fleet-transfer rationale. Escalation ladder updated with external-acquisition rung; manufacturer Field Adapter check required before local training proposals (IV.4.4). IV.5.1 restructured to three context-delivery and computation-placement modes, adding host-served operation as the enterprise default (graph and computation remain on the Host; the subordinate device captures and displays; results looped back over qBEAP™), with deterministic, logged per-session mode selection and connected/offline mode combination; projections declared provisioned content under Annex II §II.6.2 (IV.5.1); boundary statement extended to the Distribution Boundary (IV.5.2). Sync grants gain mandatory lease field (IV.5.3); roundtrips include lease renewal with revocation and version check (IV.5.4); down-sync mapped onto Annex III §4.5 provisioning classes with Organizational Authority Consent covering the confidential class, lease renewals Class 1 (IV.5.7). Sealing formalized as device-identity binding plus custody invariant (IV.5.9, IV.7.4). New IV.5.10: lease semantics for offline

Version Date**Change**

		projections — fail-closed expiry, renewal-with-checks, cumulative relationship to the custody invariant, bounded revocation latency. IV.7.6 and IV.7.8 updated for lease coverage. IV.8.3 extended: Host gating or provisioned delegated enforcement within a declared tier ceiling; offline consequential steps halt as pending confirmations. IV.6 restructured: WR Code® defined as a trigger initiating a BEAP™ exchange, in exactly two classes (public / internal) with deterministic single-path resolution and silent non-acceptance for non-enrolled scanners (IV.6.1); BEAP™ payload taxonomy made explicit — session templates, context blocks, capability declarations and requests, LoRA-class adapters — delivering the specialized AI in one secure exchange (IV.6.2); two distribution paths defined (IV.6.3): the public repository path with mandatory public auditability of all artifacts including safetensors (anti-infection control; no sealed content in the public repository) and the direct message path (pBEAP™/qBEAP™) permitting confidential inter-organizational delivery under mandatory compensating controls; internal template governance retained (IV.6.4); new IV.6.5 Mandatory Evidence Annotation for directly delivered artifacts — hash, publisher, version, capability manifest, timestamp, retained by both parties for bilateral provability, with optional anchored public hash-commitment registration (SHOULD in Silicon Cargo™ / High-Assurance). New IV.3.8 Adapter Transport in BEAP™ Packages: LoRA-class adapters as safetensors-only payload artifacts in any package class, blob-inert in transit, transport-confers-no-trust rule (ingestion, verification, Class-3 admission regardless of channel), class-dependent carriage rule (pBEAP™ adapter payloads reference repository-hosted, publicly auditable artifacts by hash — no unregistered weights over pBEAP™; direct confidential weight carriage qBEAP™-only under evidence annotation), distribution-vs-synchronization delimitation against IV.5.2, PoAC™-gated acquisition / PoAE™-gated exercise summary, Silicon Cargo™ / High-Assurance manipulation-risk note. IV.4.2 extended with Media Reconstruction and Analysis Placement: media originals are never decrypted, decoded, or persisted in the Host trust root in any tier; they are routed to the sandbox trust domain (hypervisor-isolated sandbox VM co-resident on single-device deployments — crosvm microVM / Hyper-V / VirtualBox per platform tier — dedicated sandbox hardware where the tier provides it), decoded there, and re-encoded into Reconstructed Media Artifacts — perceptually identical, structurally new, no original byte structure, metadata extracted as text; the Host AI and its perception adapters analyze reconstructions only (light conversion on sandbox hardware, heavy inference on the Host); originals remain in sandbox custody as hash-bound evidence; non-reconstructible media fails closed and never leaves the sandbox; pixel-carried text remains evidence data, never instruction; ephemeral converter instances SHOULD in High-Assurance; IV.5.6 media sync switched to reconstructions (originals not distributed). Worked example updated (lease, live Host gating, offline-halt behavior). Claimed-contribution statement updated (lease-bound, device-sealed projection; manufacturer acquisition). Canon-wide terminology migration applied: IV.6.3 direct-message-path wording revised — "hidden path" replaced by "confidential delivery path" and "exchanged in secret" by "exchanged confidentially" (no semantic change; this path is the distribution path and is distinct from the Confidential Path of the Canon's runtime data model). IV.6.1 extended with a forward pointer: a public WR Code® carrying a handshake invitation participates in the fast public initiation class of Annex IX (public_bearer); resolution-class determinism, single-path resolution, and silent non-acceptance for internal codes explicitly unaffected. Infographic IV-N3 scheduled for regeneration under the WR Preparation stage vocabulary (see
1.2	2026-07-13	

Version Date

Change

editorial note).

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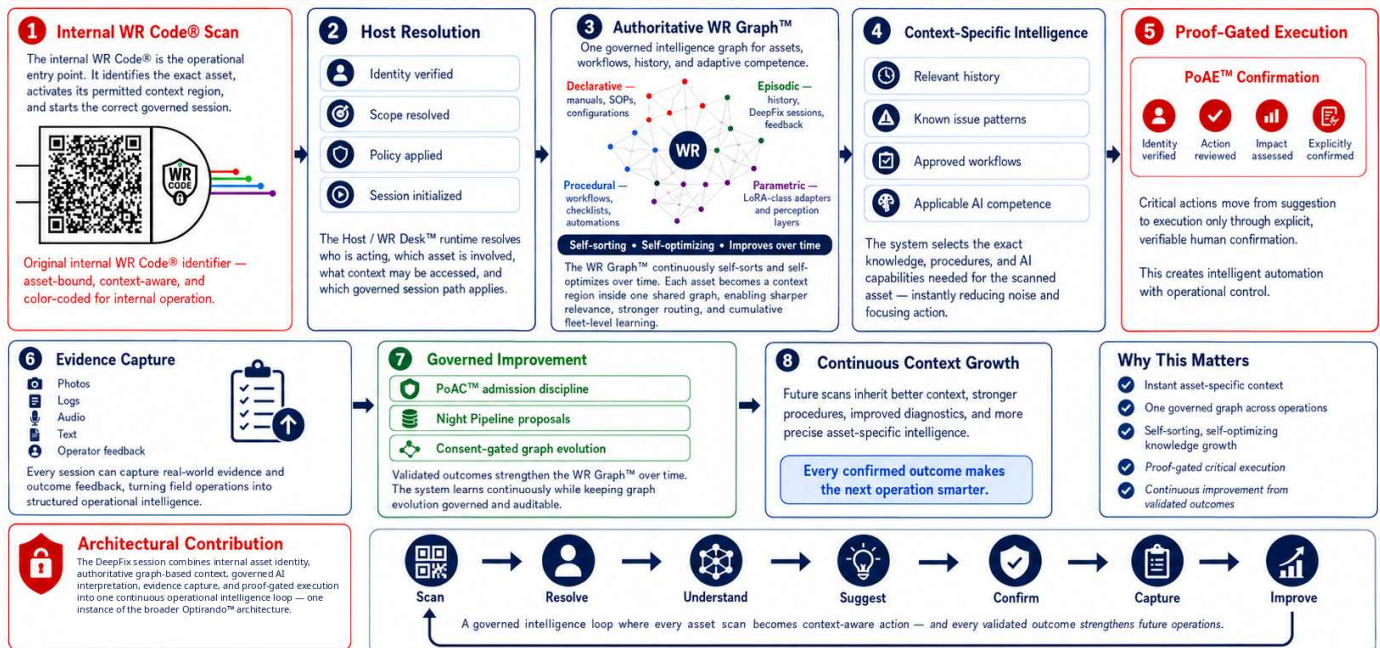
This Annex forms an integral part of the Optirando™ specification and is licensed under the same terms as BEAP™, WR Desk™, and PoAE™.

DeepFix Session — Under Internal WR Code® in Optirando™

DeepFix under WR Code® — 1/8

How a single internal WR Code® scan becomes a context-aware, governed DeepFix session

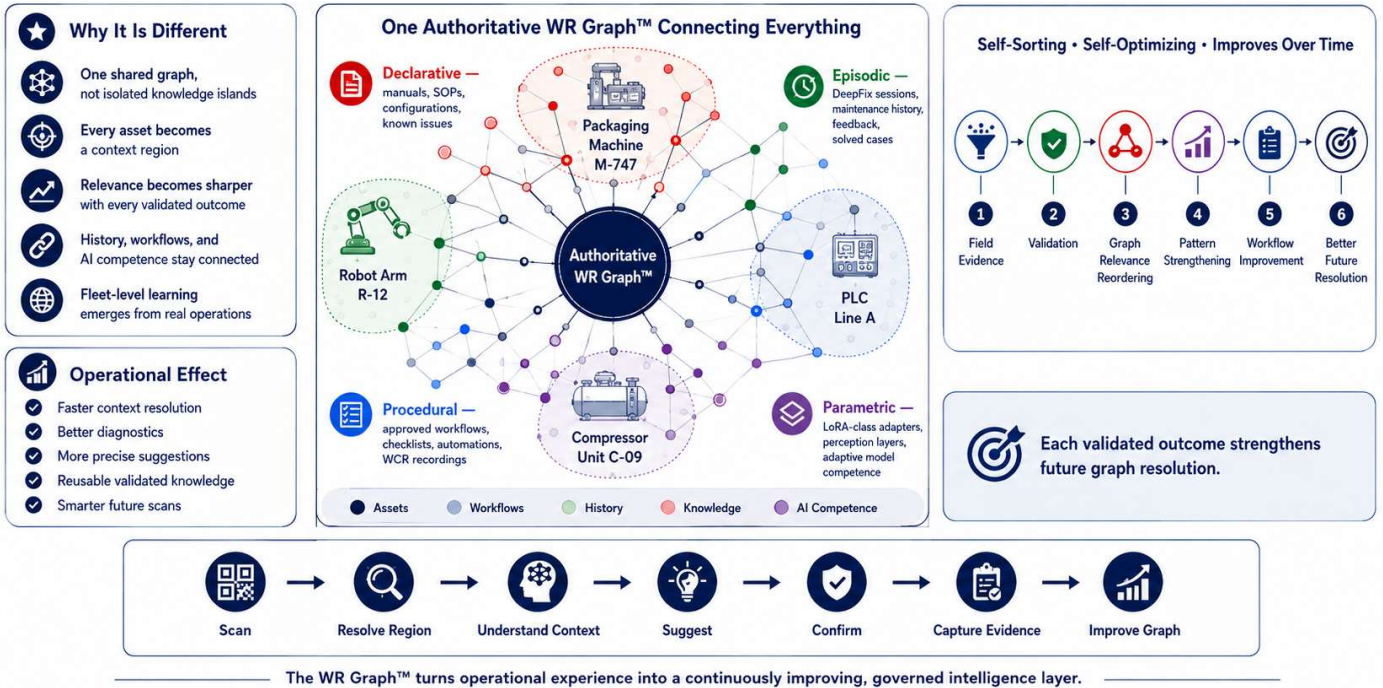
The DeepFix session: WR Desk™ turns a registered asset's WR Code® scan into an intelligent, governed operational context — combining identity, memory, workflows, AI interpretation, evidence capture, and proof-gated execution in one continuous session loop.



WR Graph™ — Self-Sorting, Self-Optimizing Intelligence Graph

DeepFix under WR Code® — 2/8

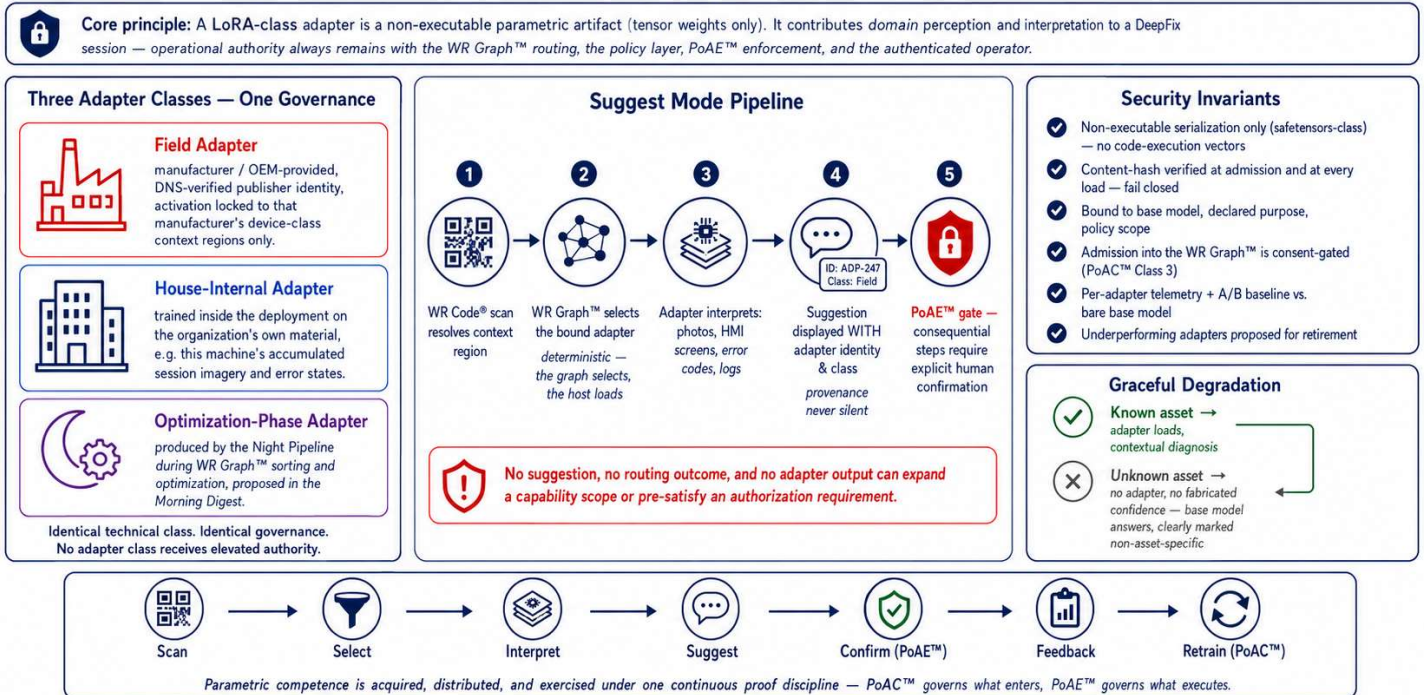
Infographic 2 of 8 • The context layer behind every DeepFix session — one authoritative graph for assets, workflows, history, and adaptive intelligence



LoRA-Class Adapters — Competence Without Authority

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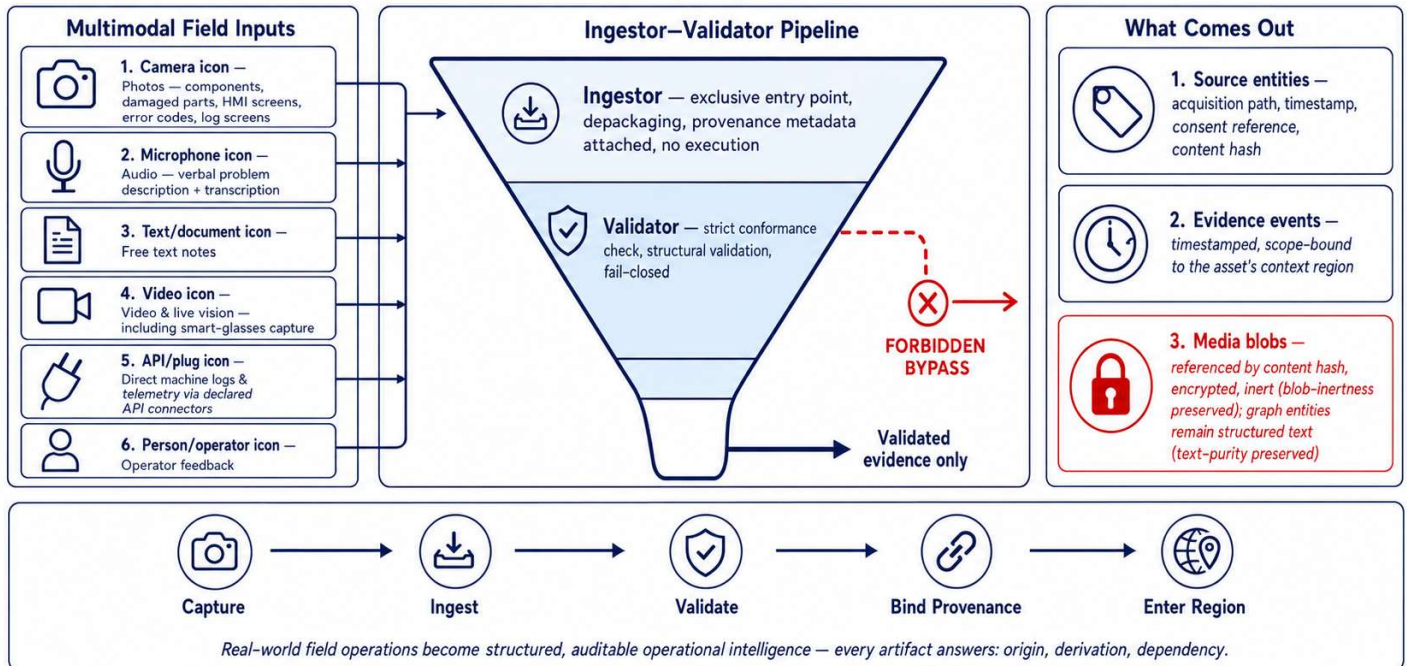
How DeepFix gains machine-specific perception under Optirando™ — adapters propose, they never decide



The DeepFix Capture Funnel — Nothing Enters Raw, Nothing Enters Anonymously

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Every field signal becomes provenance-bound evidence in the machine's WR Graph™ region

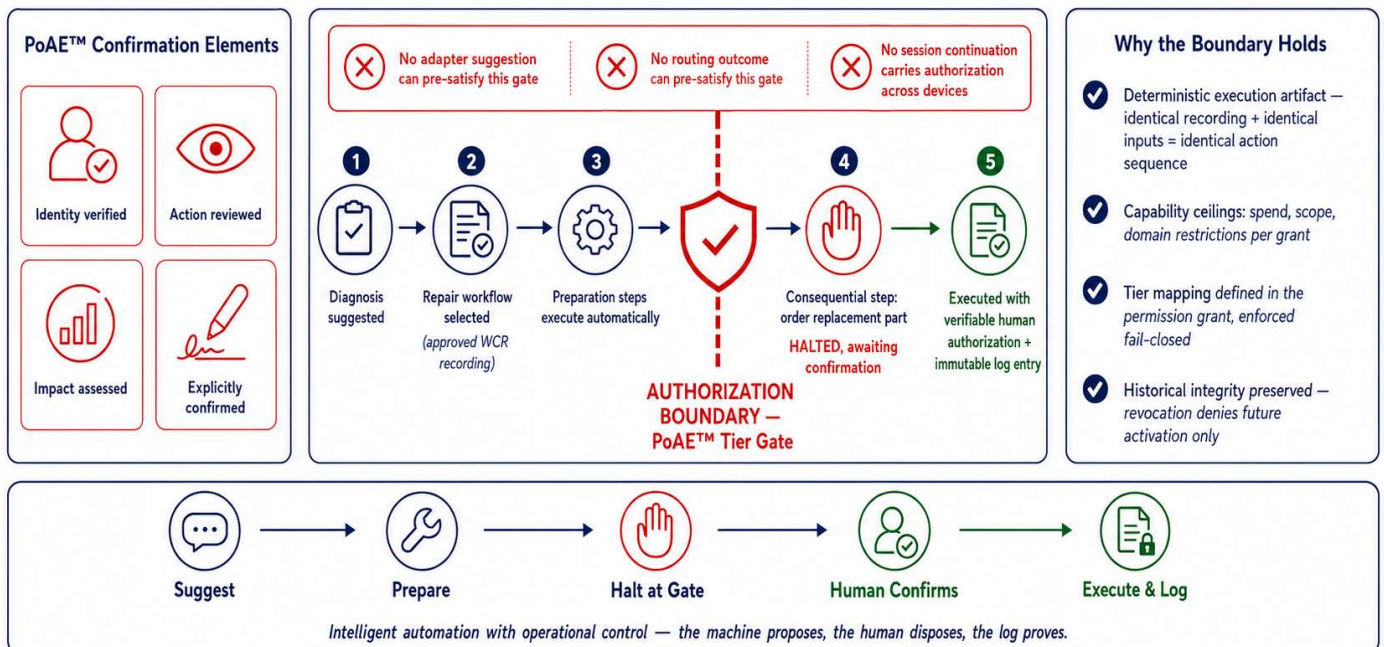


Optirando™ · WR Desk™ · DeepFix · Annex IV, 2026

Proof-Gated Execution — Automation Runs to the Boundary, and No Further

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Every consequential step in a DeepFix session halts at its PoAE™ tier until a human confirms

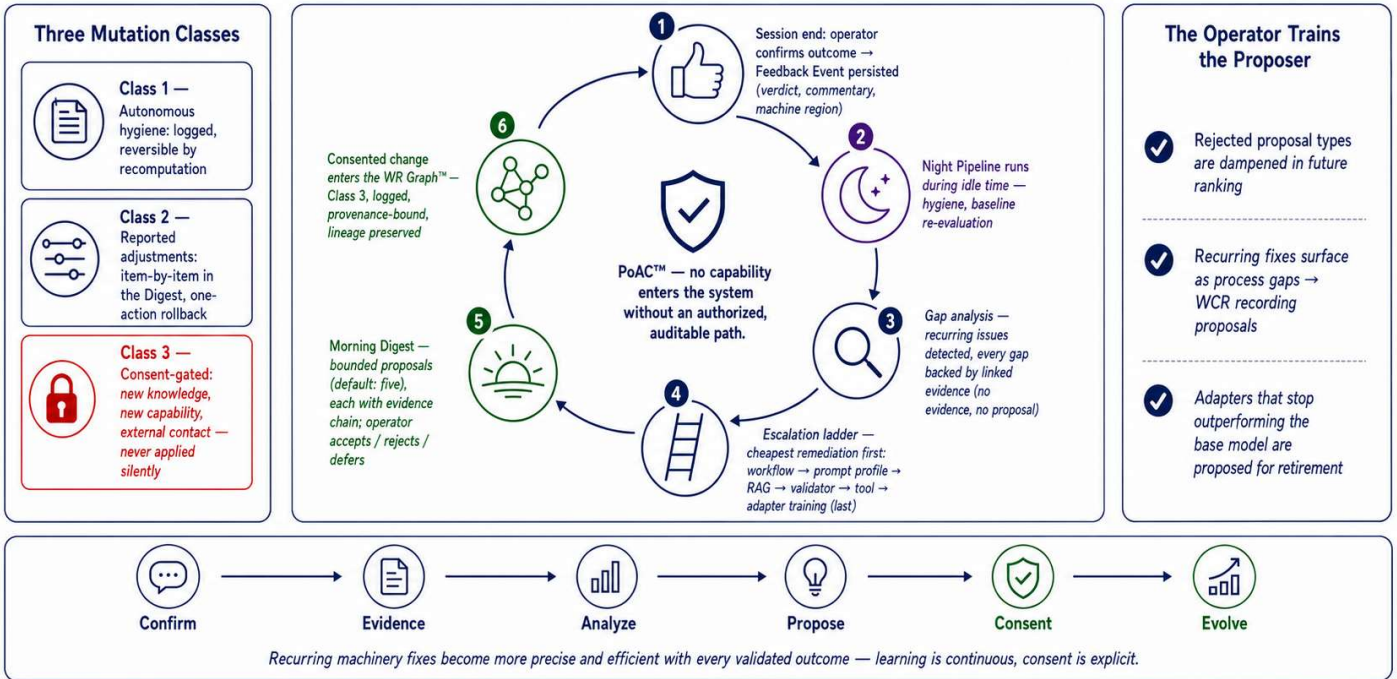


Optirando™ · WR Desk™ · PoAE™ · DeepFix · Annex IV, 2026

One Thumbs-Up Trains the Factory — Governed Evolution Under PoAC™

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Session feedback becomes evidence; evidence becomes proposals; nothing enters the WR Graph™ without consent

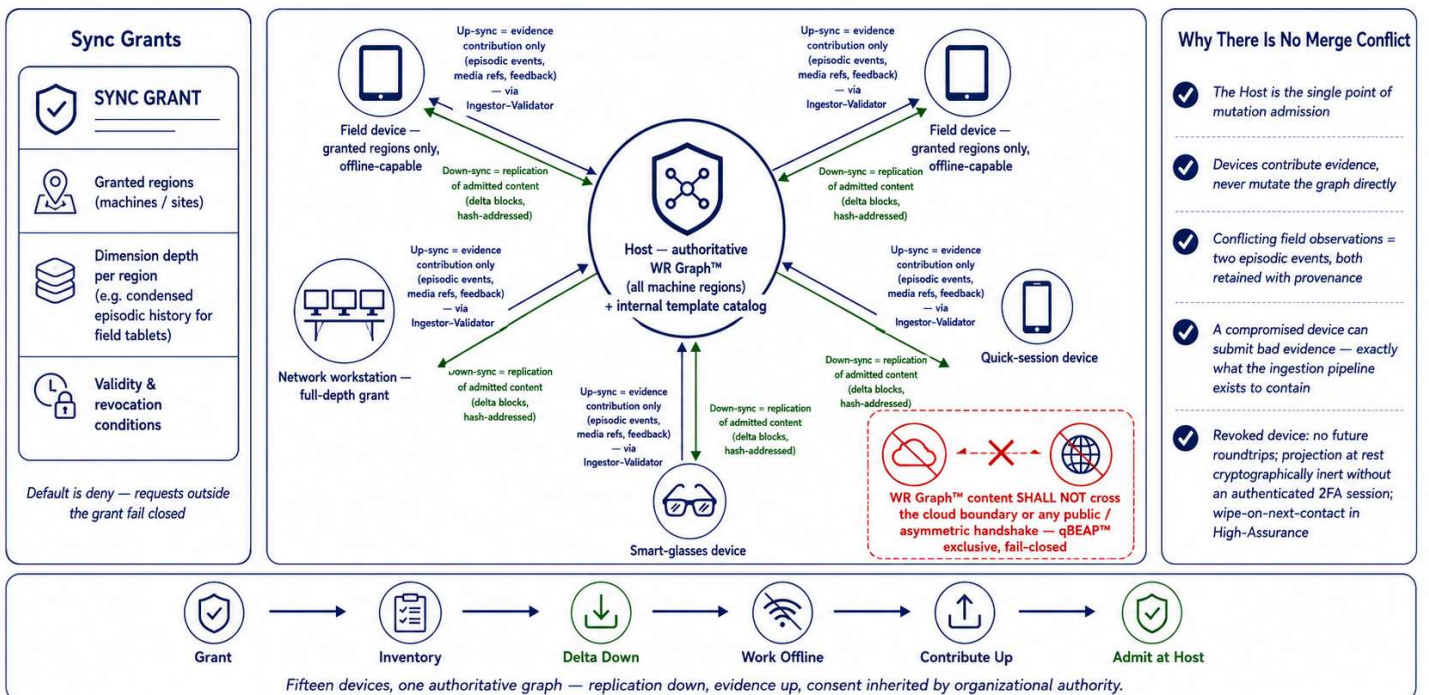


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Graph Region Synchronization — The Host Carries Everything, Devices Carry What They're Allowed

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Sealed qBEAP™ roundtrips inside one trust domain — never public, never cloud

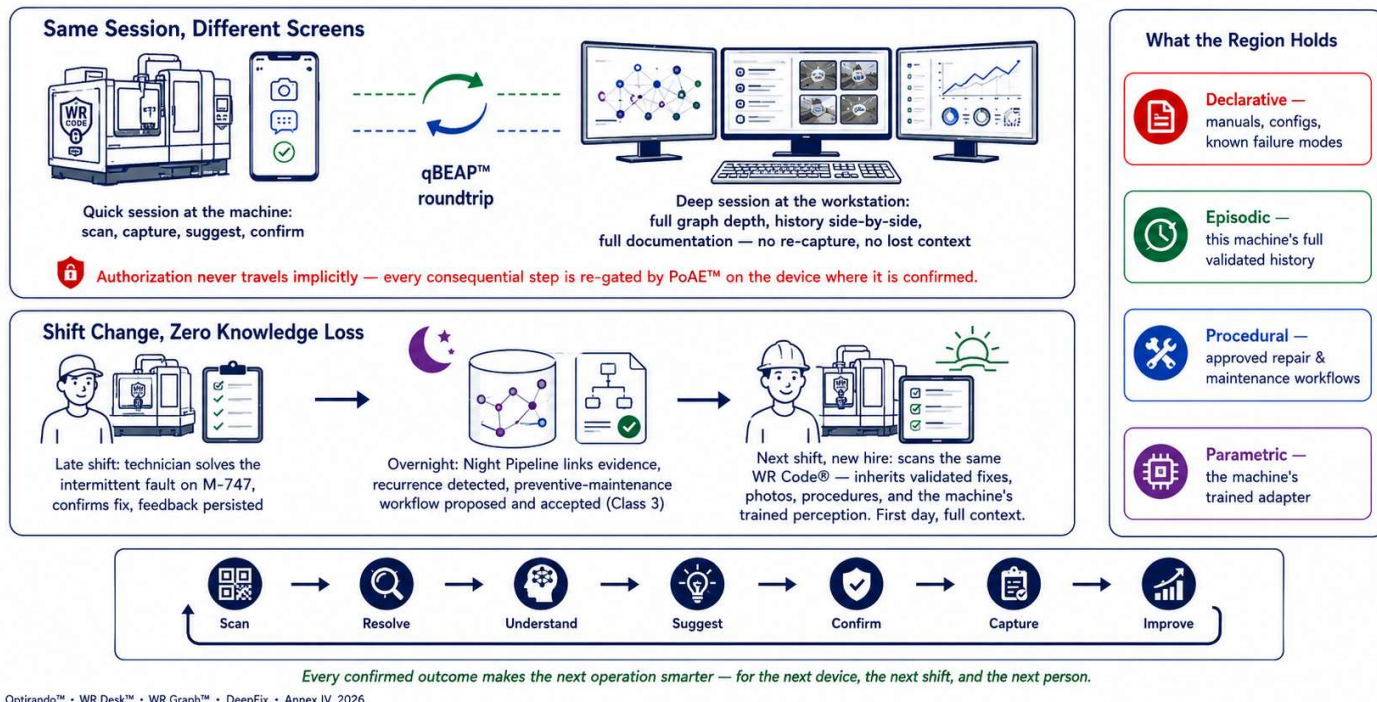


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Continuity — Across Devices, Across Shifts, Across People

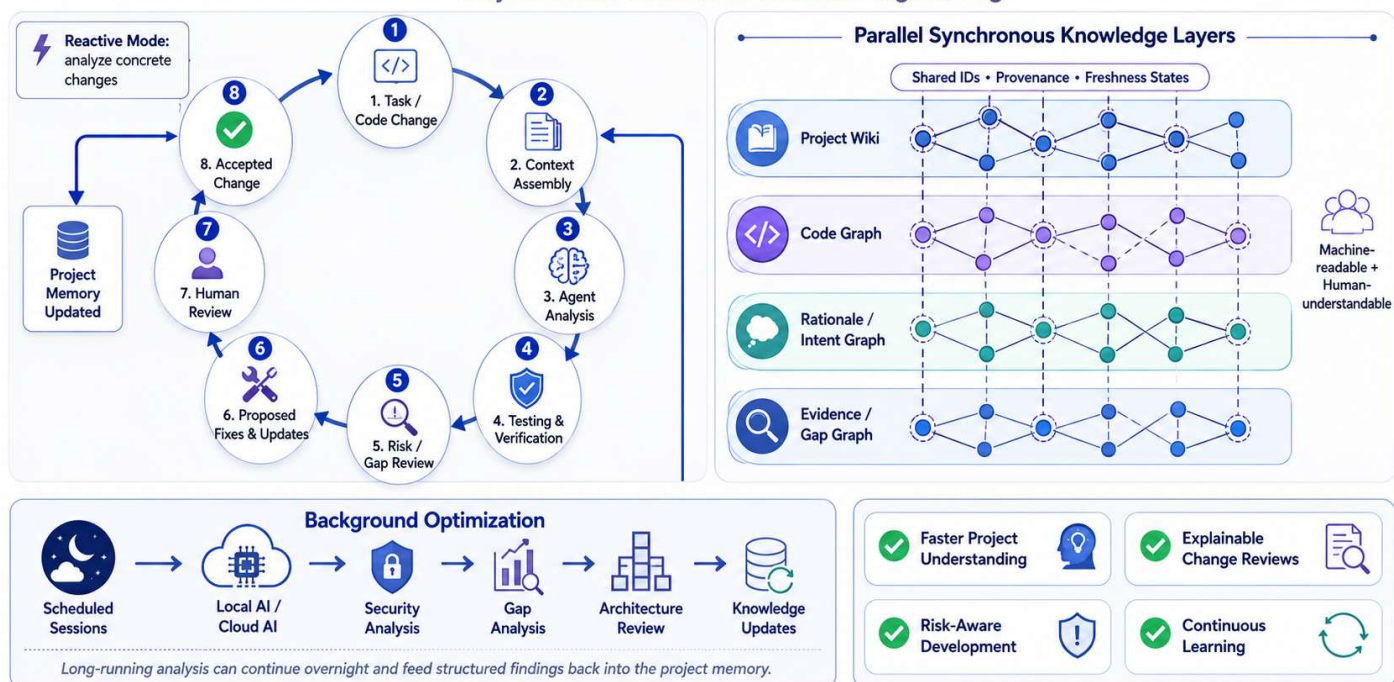
The machine's knowledge lives in its WR Graph™ region — not in someone's head

8 / 8



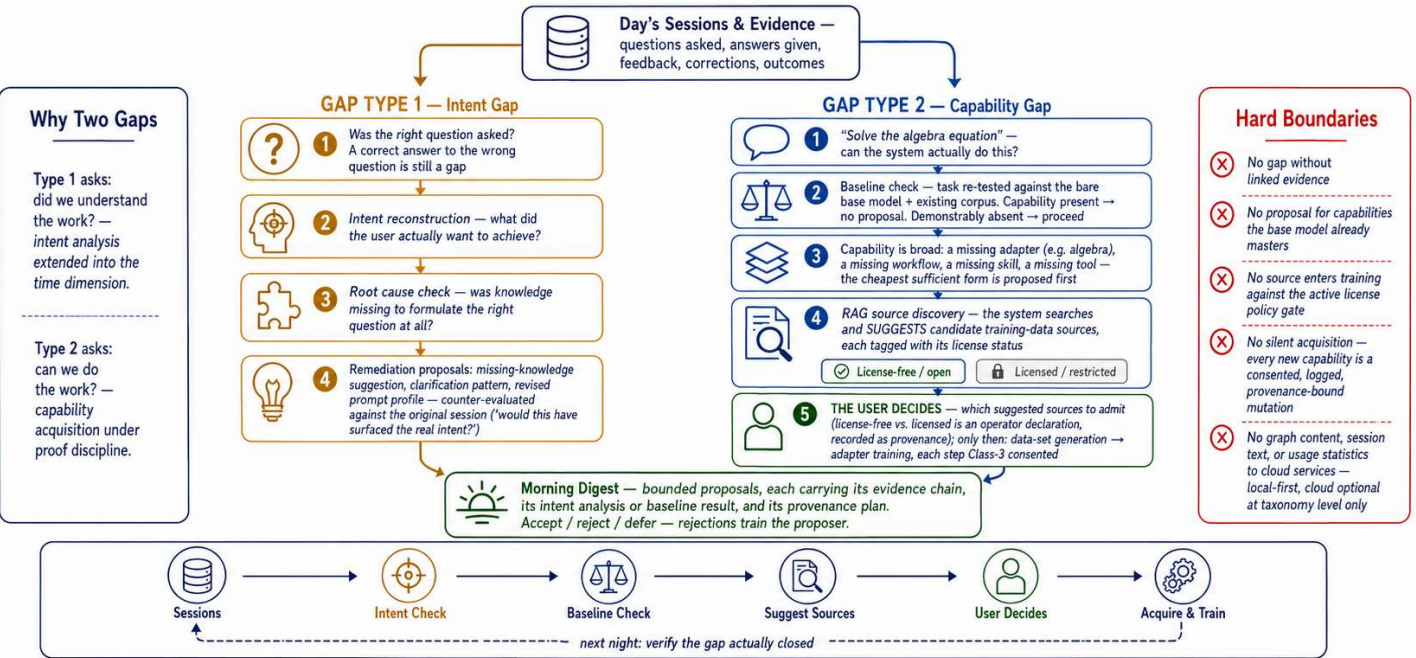
Glassview: Continuous Optimization Loop + Synchronous Project Graphs

Project-aware AI-assisted software engineering



The Gap Analyzer — Right Question First, Missing Capability Second

Nightly analysis of DeepFix sessions: reconstruct what the user really wanted, then determine what the system must acquire — under consent, with license-aware data sourcing

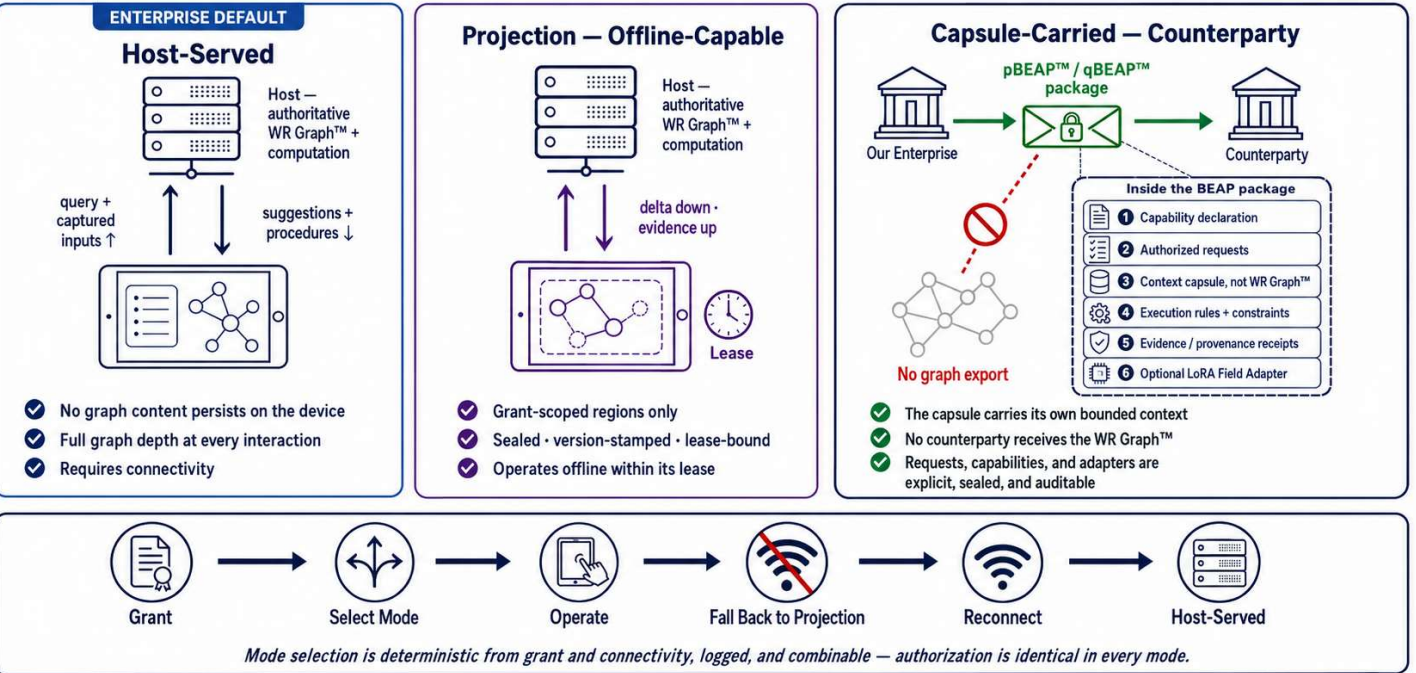


Optirando™ · WR Graph™ · PoAC™ · Night Pipeline · Annex IV, 2026

Field Operation — Three Modes, One Governance

IV-N1

Where the graph lives and where the computation runs is a deployment choice — authorization never changes · extends figures 1/8 and 7/8

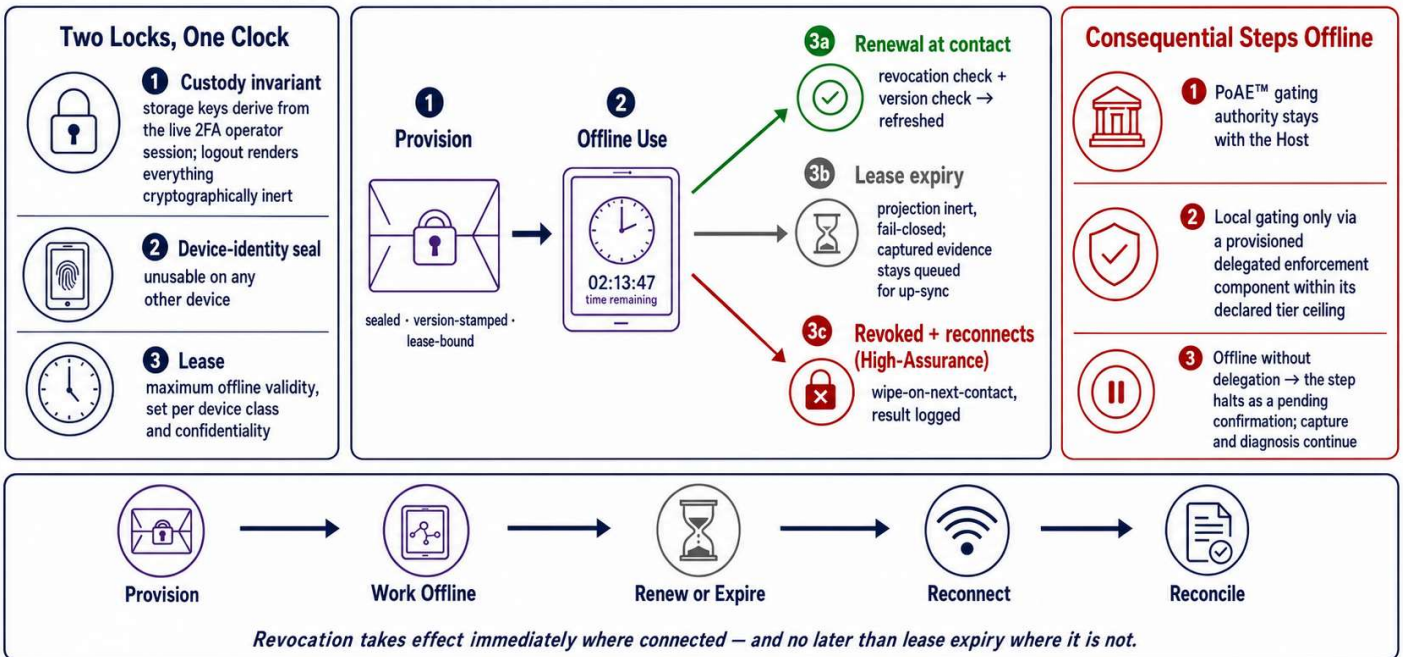


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Lease & Custody – The Offline Security Model

Two independent locks on every projection • extends figures 7/8 and 8/8

IV-N2

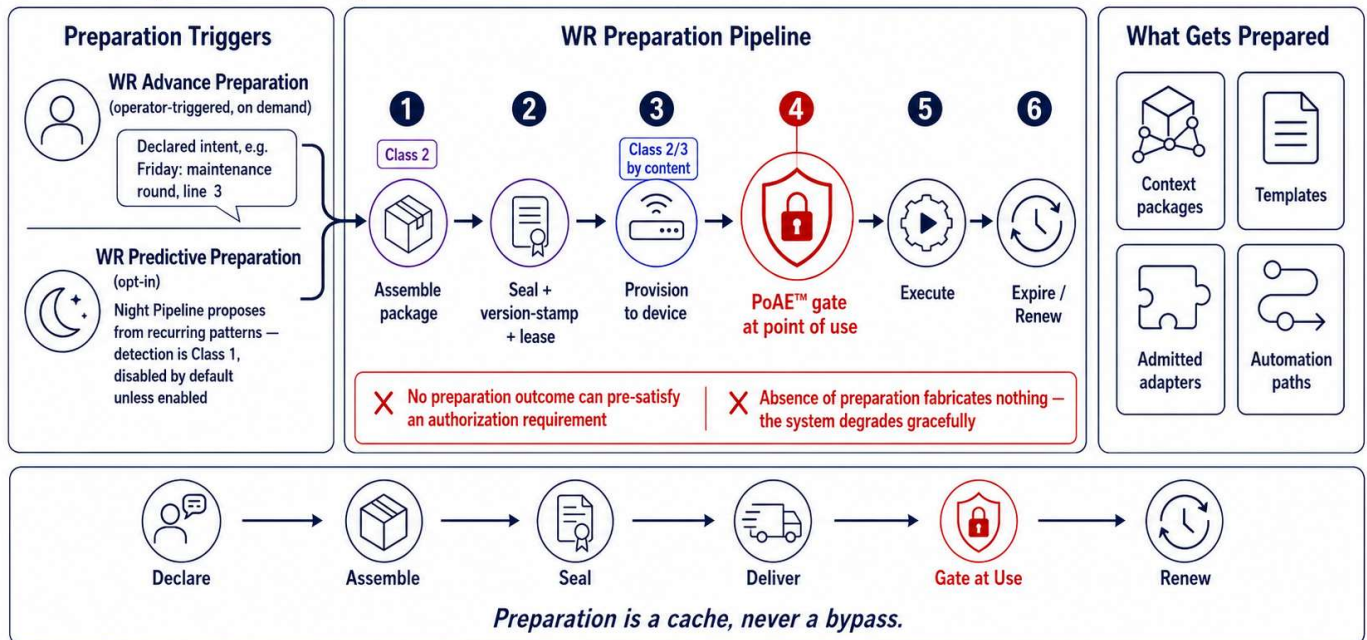


Optirando™ • WR Desk™ • qBEAP™ • Annex IV §§IV.5.9–IV.5.10, IV.8.3 • Annex II §II.3.1 • 2026

WR Advance Preparation – Ready Before It Is Needed

Idle time and storage traded for runtime independence – never for authorization

IV-N3

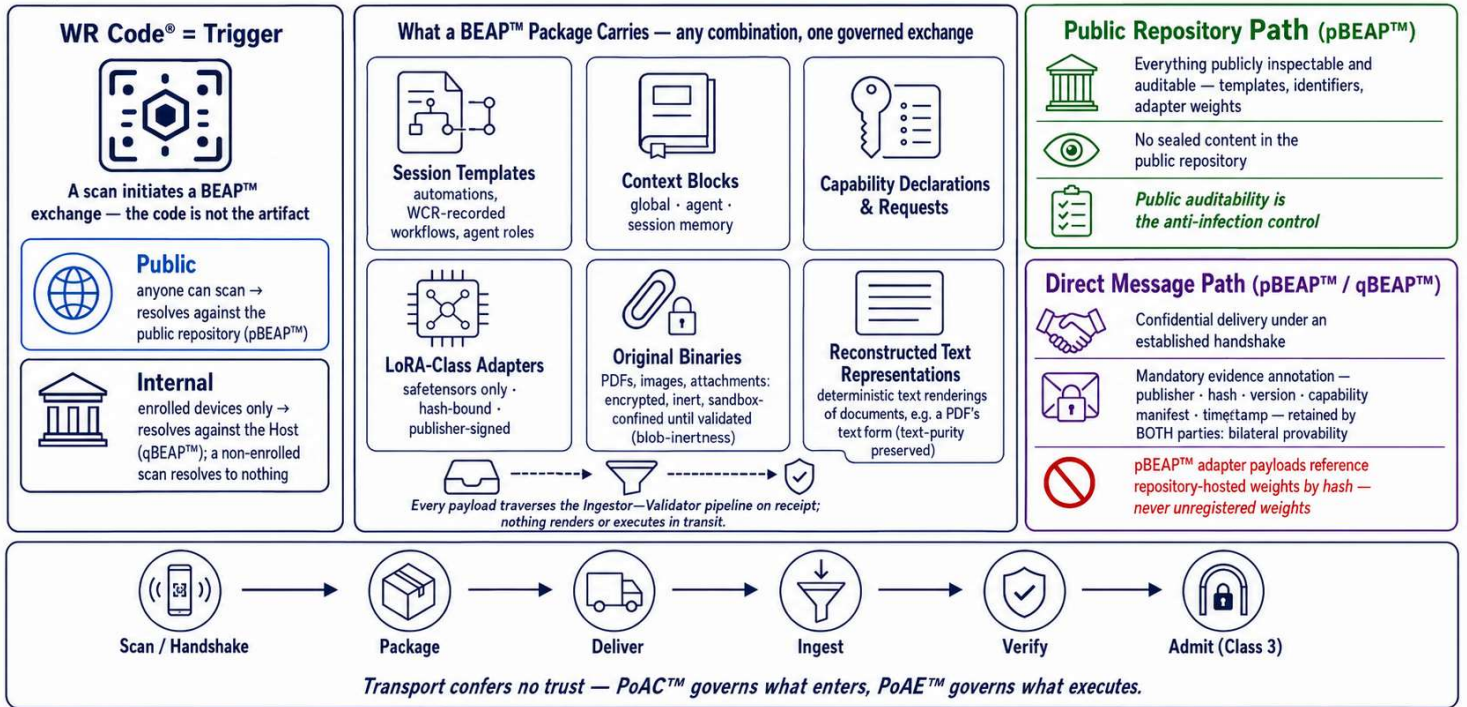


Optirando™ • WR Graph™ • PoAC™ • Annex III §§II.6.5, II.2.5 • Annex III §6.1 • 2026

One Trigger, One Package, Two Paths

The specialized AI, delivered in a secure format

IV-N4

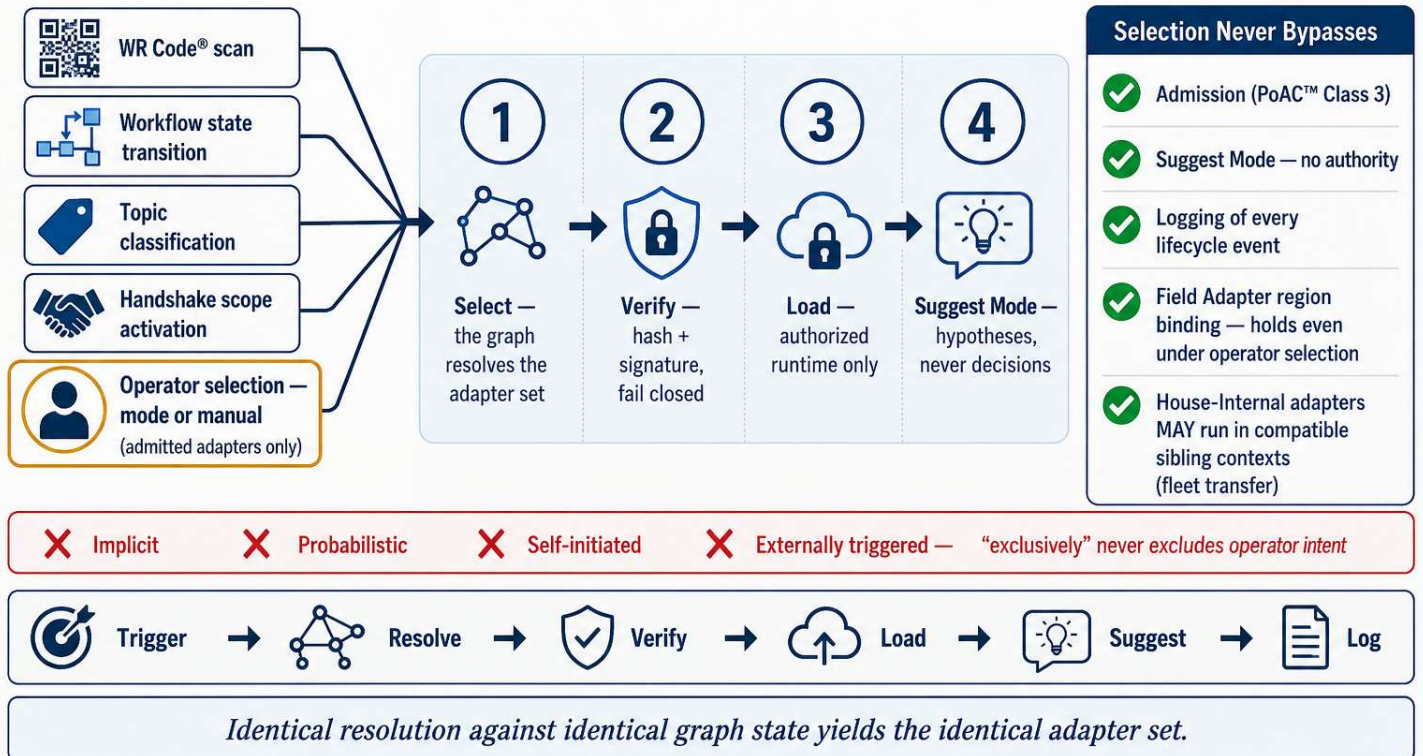


Optirando™ · WR Desk™ · BEAP™ · Annex IV §§IV.6.1–IV.6.5, IV.3.8 · 2026

IV-N5

Adapter Activation — Five Deterministic Triggers

Exclusively deterministic — and the operator is the strongest trigger of all · extends figure 3/8



IV-N6

Two Ways Competence Enters — One Proof Discipline

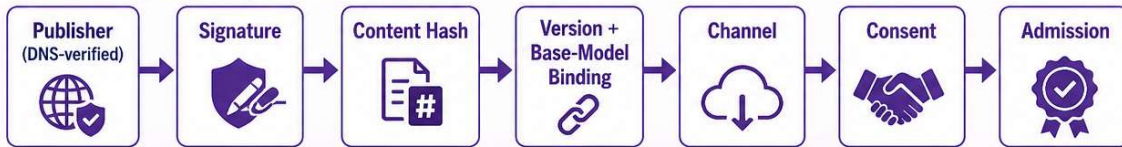
Derived and acquired adapters, and the complete remediation ladder • extends figures 3/8, 6/8 and the Deep Dive

Derived — House-Internal • Optimization-Phase



Retirement =
keep provenance,
regenerate on a
new base model

Acquired — Field Adapter (manufacturer)



Retirement =
deactivate +
optional
manufacturer
update check —
never regenerate;
installing an update
is a new acquisition
(Class 3)

The Remediation Escalation Ladder

Exhaust lower rungs
before climbing higher.



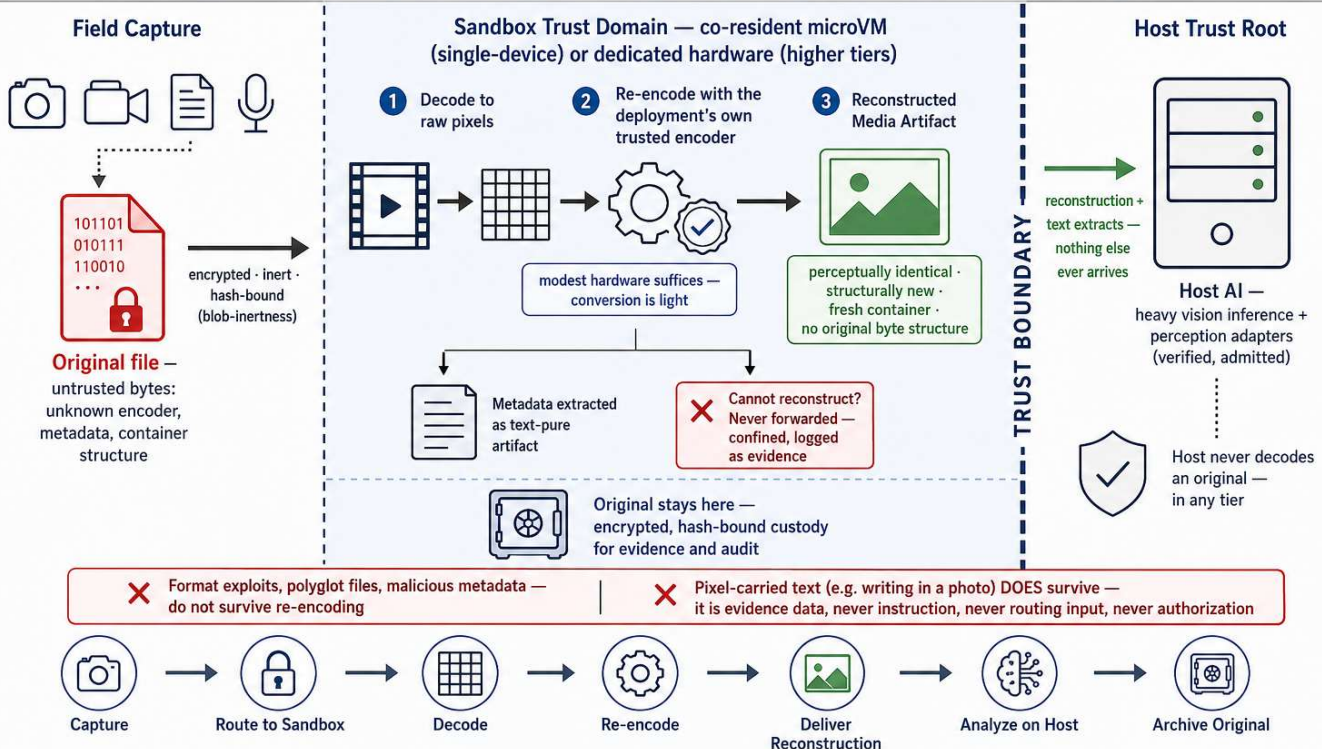
Cheapest rung first — check the manufacturer before training your own.

Optirando™ • PoAC™ • Annex III §§3.4, 3.8, 6.2, 6.3 • Annex IV §§IV.3.1, IV.4.4 • 2026

Media Reconstruction — Only the Pixels Cross, Never the Bytes

Original files never reach the Host trust root — the sandbox converts, the Host understands • extends figure 4/8

IV-N8



Light conversion on weak hardware, heavy inference on strong hardware — security and performance from the same boundary.

Optirando™ • WR Desk™ • WR Graph™ • Annex IV §IV.4.2 • 2026

Closing Note (Informative)

This Annex, in its revised form, specifies the complete industrial field operation of the WR Graph™: one authoritative, region-structured graph on the Host; three modes of field operation selected by deployment rather than dictated by architecture; parametric competence that is acquired, transported, provisioned, and exercised under a single uninterrupted proof discipline; and a device, identity, and lease model under which nothing usable outlives its authorization.

The mechanisms are deliberately few, and they compose rather than multiply. Every artifact — template, context, capability declaration, adapter, binary, or reconstructed representation — enters through the same ingestion pipeline, carries the same class of evidence, and is admitted under the same consent taxonomy, whether it arrives from a public scan, a confidential inter-organizational message, a manufacturer, or the deployment's own Night Pipeline. Every suggestion, however it was routed, prepared, or transported, stops at the same gate.

The placement principle extends to perception itself. Field media never reach the Host trust root as original files: they are routed to the sandbox trust domain — the hypervisor-isolated sandbox VM co-resident on a single-device deployment, using the platform's isolation primitive, dedicated sandbox hardware where the tier provides it — decoded there, and re-encoded into reconstructions that are perceptually identical and structurally new. Only the pixels cross the trust boundary, never the bytes. The Host AI performs its heavy vision inference exclusively over reconstructions; the originals remain in sandbox custody as hash-bound evidence; format-level exploits, polyglot files, and malicious metadata do not survive the re-encoding, while text visible within imagery — which does survive, by design — remains evidence data, never instruction.

The mechanism is not specific to field capture: it applies identically to every media file the system receives, whether captured at a machine, carried in a depackaged email, or delivered in a direct BEAP™ message. Attachments and message content — PDFs, images, videos, the email's original visual design — are reconstructed in the sandbox and displayed on the Host without perceptible difference: the operator reads the PDF, watches the video, sees the message exactly as its sender designed it, and at no point has an original byte touched the trust root. An earlier revision of this architecture deliberately refrained from rendering such content at all, because the rasterizer or conversion tool is itself an attack surface; with conversion confined behind a hypervisor boundary — ephemeral where the platform supports it, resettable, structurally contained — that objection dissolves. What this yields is not merely a security property but the decisive usability property of the system: the operator uses it entirely normally, viewing what appear to be the originals, while the originals themselves never leave custody. Blob-inertness thereby holds without a single exception; the same reasoning that isolates depackaging isolates understanding, and now, presentation. The normative anchoring of reconstruction-based display for the general depackaging path belongs to the Canon and is recorded here as a direct consequence of this Annex's mechanism.

What differs between deployments is therefore placement in every sense — where knowledge resides, where computation runs, where perception decodes, what a device may hold and for how long — and placement is a governed, logged, reversible decision, never a trust decision made by default. The figures of both series depict one and the same system at different depths: where a reader finds a mechanism in a figure, its normative definition is in the referenced section; where a reader finds a rule in the text, its operational shape is in a figure. The specification claims no

component as novel in isolation. It claims the composition: capability acquisition, capability distribution, capability placement, and capability exercise under one continuous, auditable proof discipline — from the first scan at a machine, decoded at arm's length from the trust root, to the last adapter retired against its own telemetry.

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